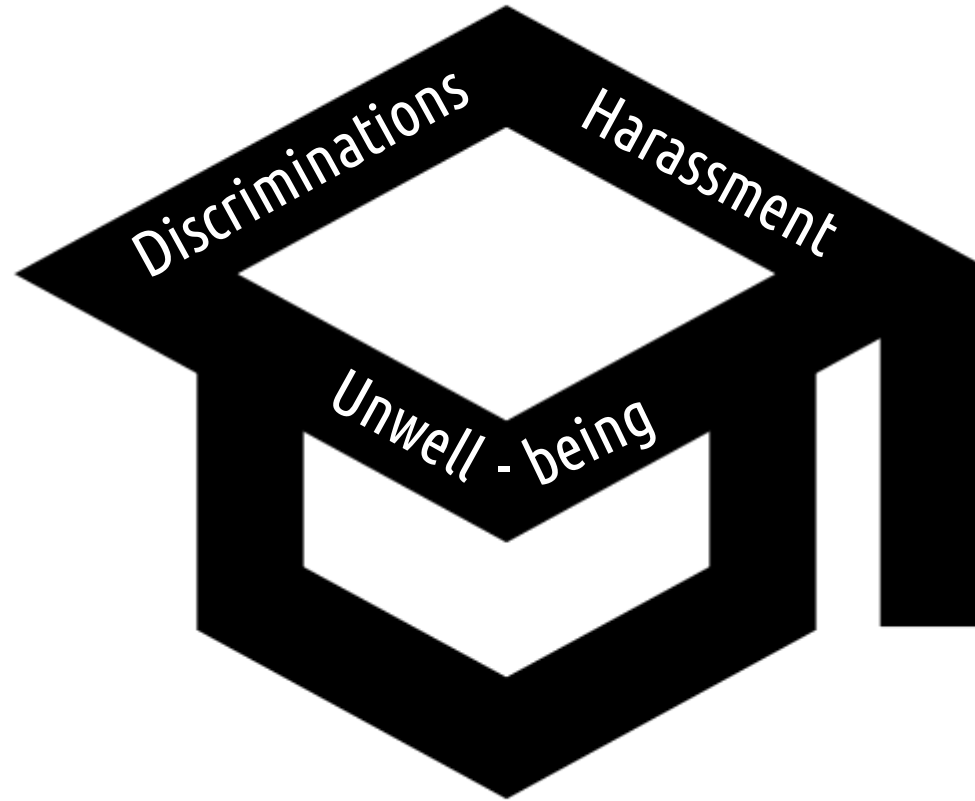




Understand
Prevent
React



Understand
Prevent
React



BUT...



BUT...

How to do good research?

We are researchers!

Introduction



BUT...

We are researchers!

A problem? → Hypothesis

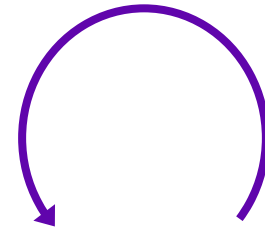


BUT...

We are researchers!

Be wrong, do better!

A problem?



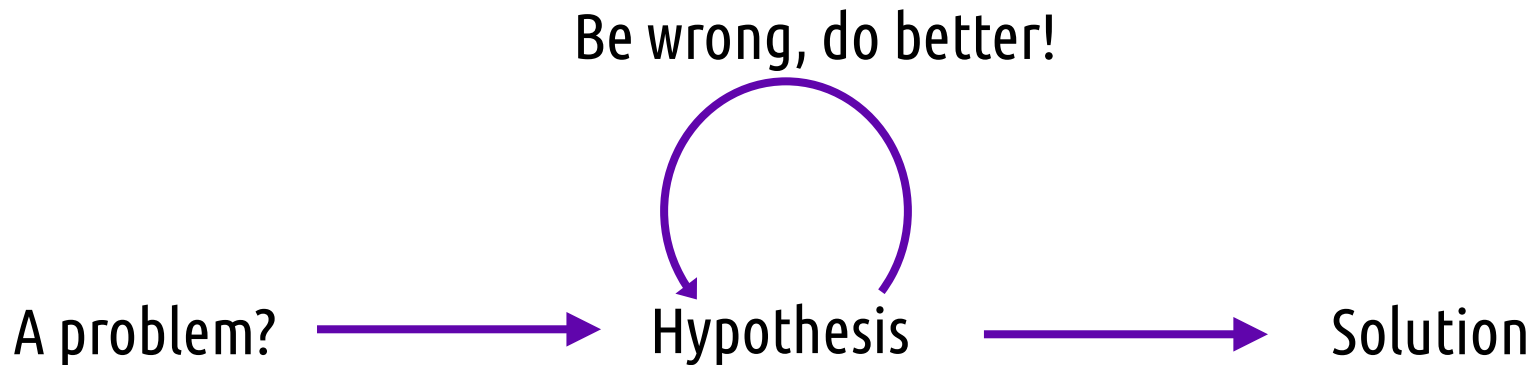
Hypothesis



How to do good research?

BUT...

We are researchers!



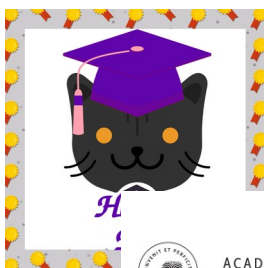
Introduction

How to do good research?

We are researchers?



Introduction



ACADÉMIE
DES SCIENCES
INSTITUT DE FRANCE



Rapport de l'Académie des sciences
Sciences : où sont les femmes ?



Report on the National Survey
Student Experiences of Sexual
Violence and Harassment in Irish
Higher Education Institutions
Summary of Survey Findings

Authors | Pádraig MacNeela, Kate Dawson,
Theresa O'Rourke, Siobhán Healy-Cullen,
Lorraine Burke, William F. Flack



How to do good research?



Introduction



Lots of sources, from reports and research papers:

- <https://not-in-our-labs.github.io/resources.html>
- <https://not-in-our-labs.github.io/presentation.html>



Lots of sources, from reports and research papers:

- <https://not-in-our-labs.github.io/resources.html>
- <https://not-in-our-labs.github.io/presentation.html>

Feedback welcome!

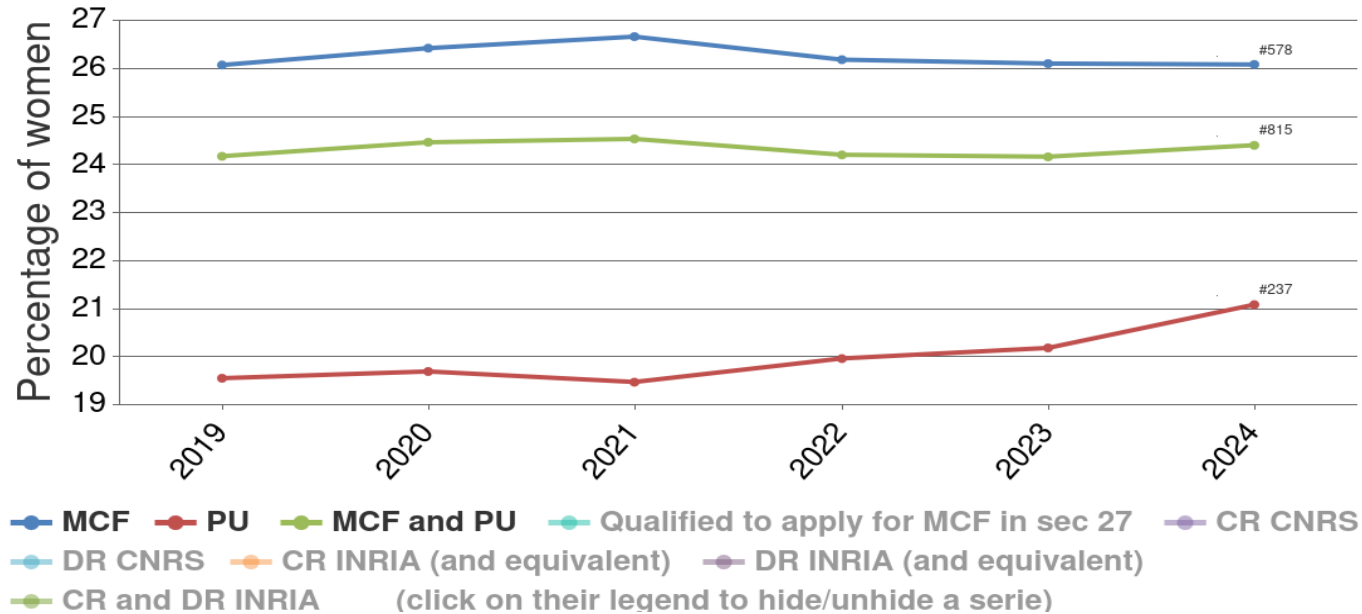


Investigation - part 1

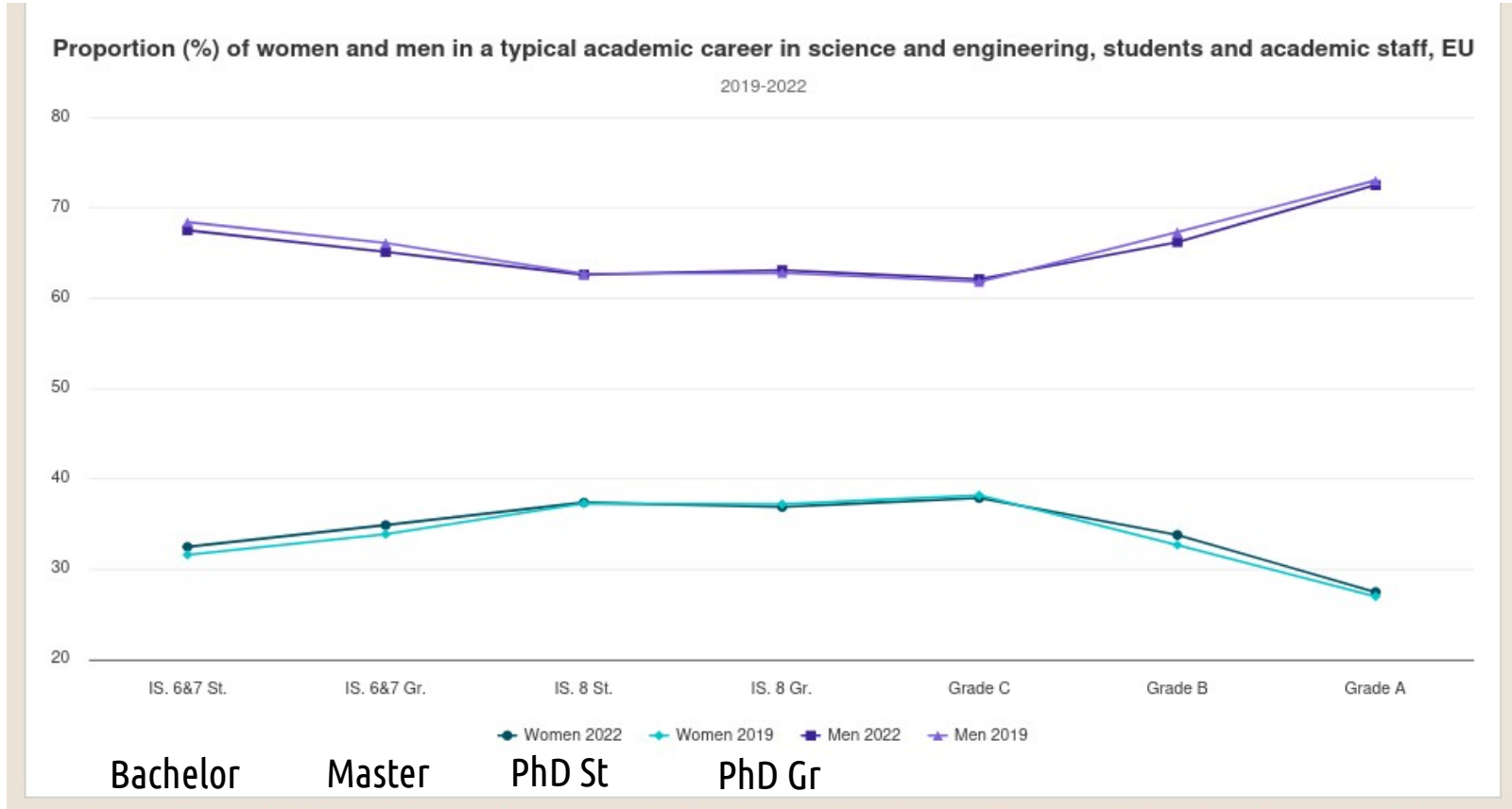


Evidence 1: a stable lack of women in CS

Proportion of women in Computer Science French academics (section 27)



Evidence 2: a leaky pipeline the higher we go



Hypothesis 1: stereotypes and biases?



Parity in computer science?



Hypothesis 1: stereotypes and biases?



Big lying belief sponge!

Hypothesis 1: stereotypes and biases?



Big lying belief sponge!

Creates a « coherent »
view of the world
from incomplete facts

Hypothesis 1: stereotypes and biases?



Change Blindness (using flicker)

(from J. Kevin O'Regan -- <http://nivea.psych.univ-paris5.fr>)

Parity in computer science?

Hypothesis 1: stereotypes and biases?

“by age 6, children already think that boys are better than girls at computer science and engineering” (Miller et al. 2024)



Hypothesis 1: stereotypes and biases?

“by age 6, children already think that boys are better than girls at computer science and engineering” (Miller et al. 2024)



Laura
a trouvé le poste de ses rêves.

C'est l'avenir qu'elle a toujours envisagé. Et l'avenir, pour elle, c'est de faire vivre et partager sa passion, transmettre des savoirs et des valeurs, se consacrer à la réussite de chacun de ses élèves. C'est pour cela qu'elle a décidé de devenir enseignante.

Julien
a trouvé un poste
à la hauteur de ses ambitions.

C'est la concrétisation de son projet professionnel. Et ce projet, pour lui, c'est de faire vivre et partager sa passion, transmettre des savoirs et des valeurs, se consacrer à la réussite de chacun de ses élèves. C'est pour cela qu'il a décidé de devenir enseignant.

L'ÉDUCATION NATIONALE RECRUTE 17 000 PERSONNES

Pourquoi pas vous ? 17 000 postes d'enseignants, d'infirmières et de médecins scolaires sont à pourvoir en 2011.

RENSEIGNEMENTS ET INSCRIPTIONS DU 31 MAI AU 12 JUILLET 2011
SUR WWW.LEDUCATIONRECRUTE.FR

L'ÉDUCATION NATIONALE RECRUTE 17 000 PERSONNES

Pourquoi pas vous ? 17 000 postes d'enseignants, d'infirmières et de médecins scolaires sont à pourvoir en 2011.

RENSEIGNEMENTS ET INSCRIPTIONS DU 31 MAI AU 12 JUILLET 2011
SUR WWW.LEDUCATIONRECRUTE.FR

Hypothesis 1: stereotypes and biases?

“by age 6, children already think that boys are better than girls at computer science and engineering” (Miller et al. 2024)



L'ÉDUCATION NATIONALE RECRUTE 17 000 PERSONNES
Pourquoi pas vous ? 17 000 postes d'enseignants, d'infirmières et de médecins scolaires sont à pourvoir en 2011.
RENSEIGNEMENTS ET INSCRIPTIONS DU 31 MAI AU 12 JUILLET 2011
SUR WWW.LEDUCATIONRECRUTE.FR



L'ÉDUCATION NATIONALE RECRUTE 17 000 PERSONNES
Pourquoi pas vous ? 17 000 postes d'enseignants, d'infirmières et de médecins scolaires sont à pourvoir en 2011.
RENSEIGNEMENTS ET INSCRIPTIONS DU 31 MAI AU 12 JUILLET 2011
SUR WWW.LEDUCATIONRECRUTE.FR



Devenir professeur des écoles

Master MEEF 1^{er} degré à l'Inspé
2022/2023



pages.univ-nantes.fr



Devenir professeur de collège ou de lycée

Master Métiers de l'Enseignement de l'Éducation et de la Formation 2nd degré à l'Inspé
2022/2023



pages.univ-nantes.fr



How to do good research?

Hypothesis 1: stereotypes and biases?

“by age 6, children already think that boys are better than girls at computer science and engineering” (Miller et al. 2024)



Parity in computer science?

Hypothesis 1: stereotypes and biases ?

So, let's just solve the gendered stereotype of science. Job's done.



Hypothesis 1: stereotypes and biases ?

So, let's just solve the gendered stereotype of science. Job's done.



The End



Hypothesis 1: stereotypes and biases ?

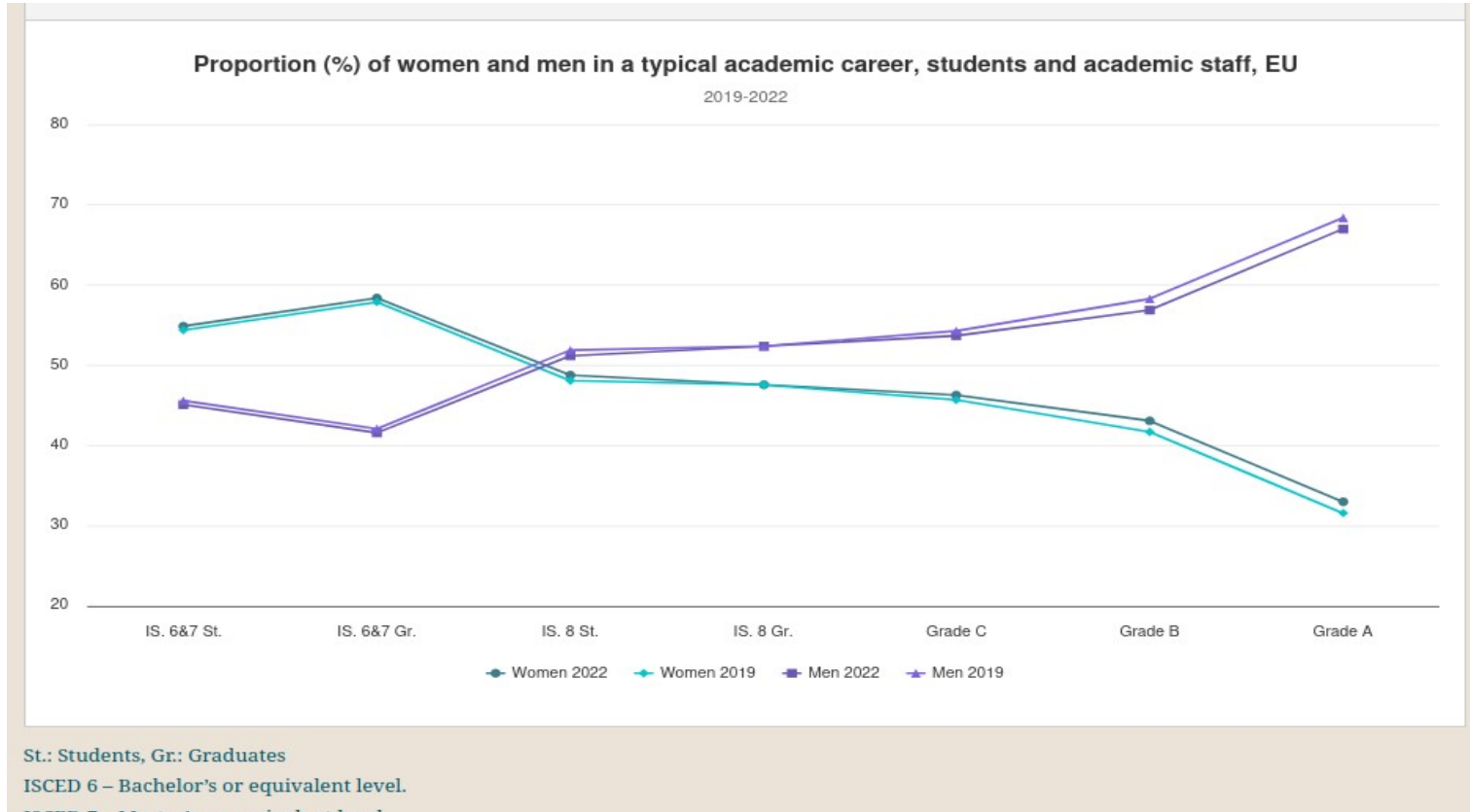
So, let's just solve the gendered stereotype of science. Job's done.



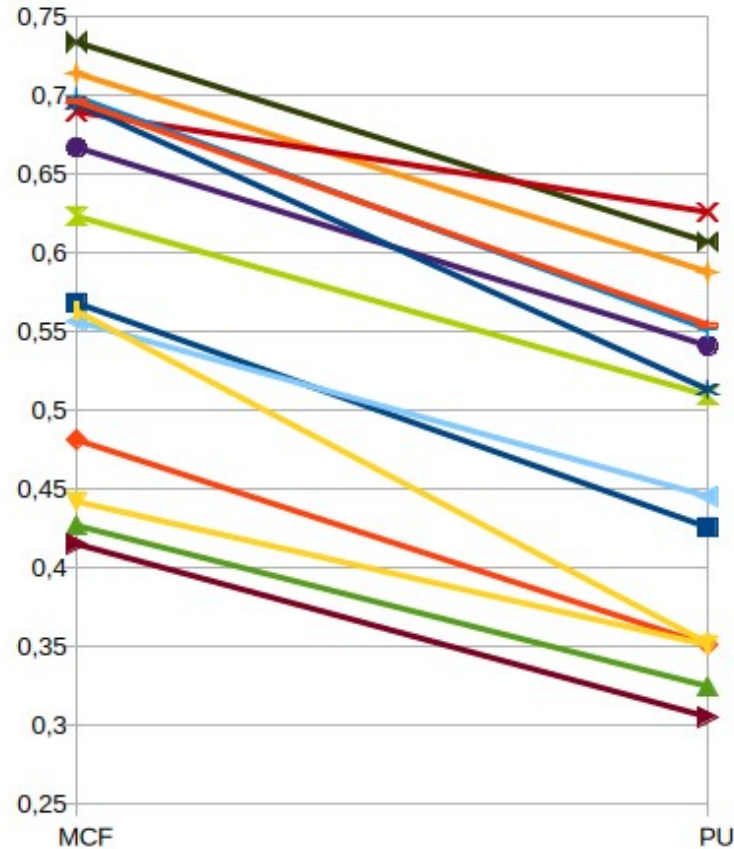
The End

*Thank
you!*

Counter-evidence: the leaky pipeline is everywhere, not just CS



Counter-evidence: the leaky pipeline is everywhere, not just CS



Share of women at the MCF and PU level in France, 2024

Right, Political sciences, language and literature

- Section 1 - Droit privé et sciences criminelles - (1870 total people)
- Section 2 - Droit public (1324 total people)
- Section 3 - Histoire du droit et des institutions (272 total people)
- Section 4 - Science politique (448 total people)
- Section 5 - Sciences économiques (1741 total people)
- Section 6 - Science de gestion et du management - (2246 total people)
- Section 07 - Sciences du langage - (767 total people)
- Section 08 - Langues et littératures anciennes (272 total people)
- Section 09 - Langue et littérature française - (823 total people)
- Section 10 - Littératures comparées (204 total people)
- Section 11 - Études anglophones (1568 total people)
- Section 12 - Études germaniques et scandinaves (323 total people)
- Section 13 - Études slaves et baltes (119 total people)
- Section 14 - Études romanes (905 total people)
- Section 15 - Langues, littératures et culture africaines, asiatiques et d'autres aires linguistiques (421 total people)



We need
more facts!



Investigation - part 2

Evidence 3: PhD students are way too depressed



*Worldwide, we have 24% of depression
among PhD students (Satinsky et al. 2021)*

- Consistently above :
- general population (~5%)
 - young adults (~15%)



How to do good research?

Evidence 3: PhD students are way too depressed

*Wordwide, we have 24% of depression
among PhD students (Satinsky et al. 2021)*

Consistently above :

- general population (~5%)
- young adults (~15%)

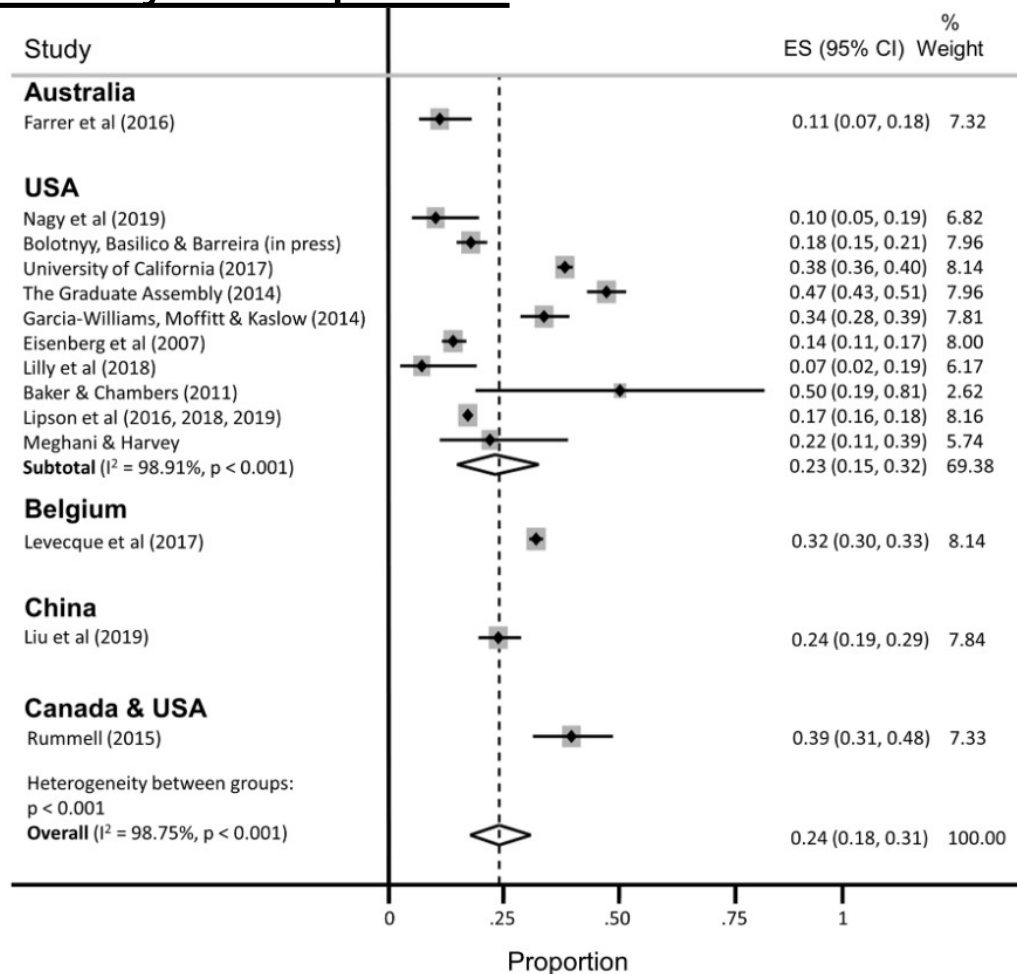


Figure 2. Pooled estimate of the proportion of Ph.D. students with clinically significant symptoms of depression

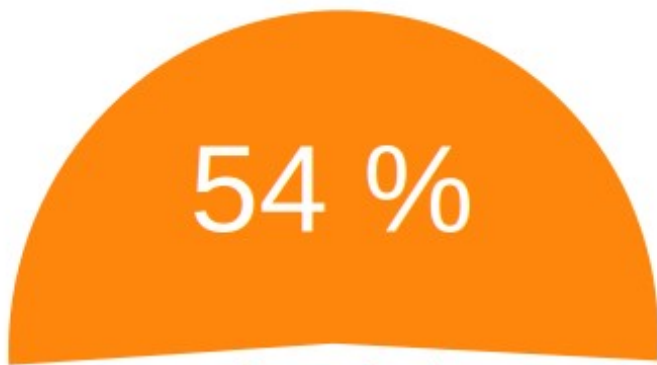
Unwell-being in academia?



How to do good research?

Evidence 3: PhD students are way too depressed

Wordwide, we have 24% of depression among PhD students (Satinsky et al. 2021)



of French PhD doctorates suffer from stress

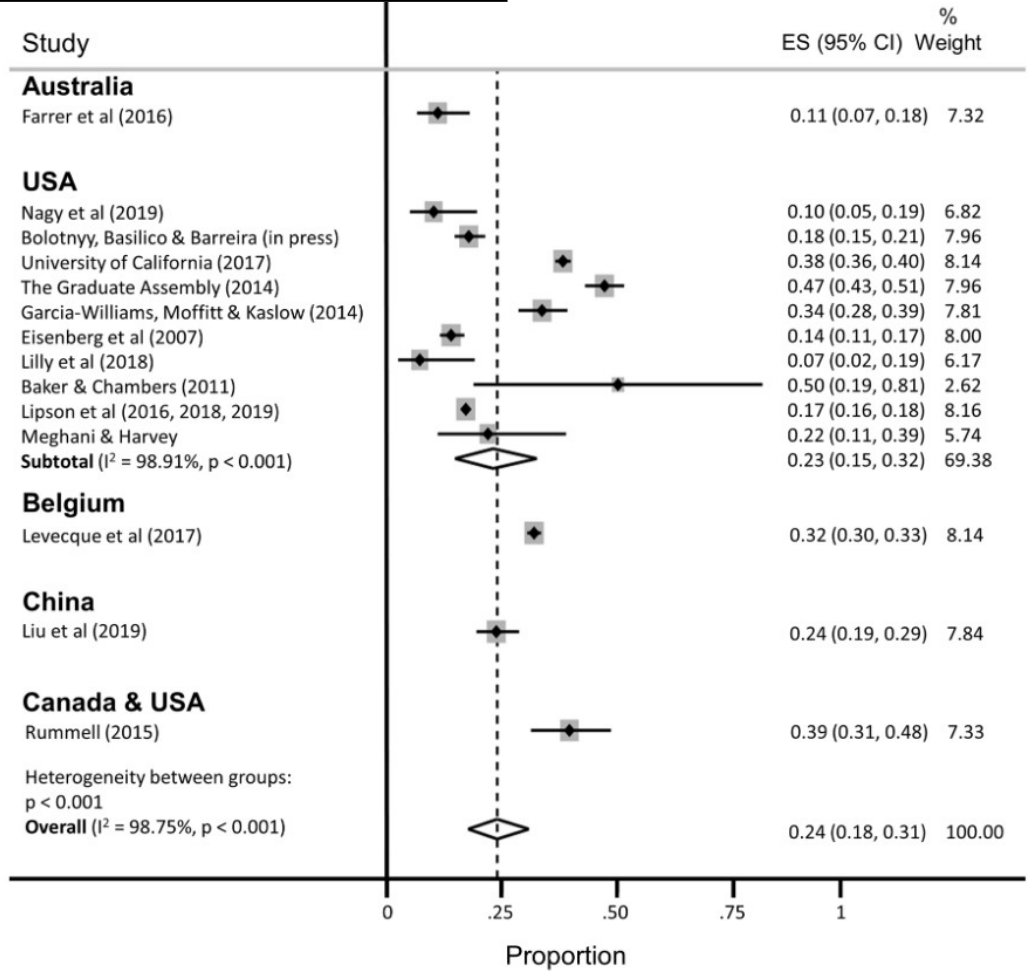


Figure 2. Pooled estimate of the proportion of Ph.D. students with clinically significant symptoms of depression

Unwell-being in academia?

Evidence 4: PhD students are overworked



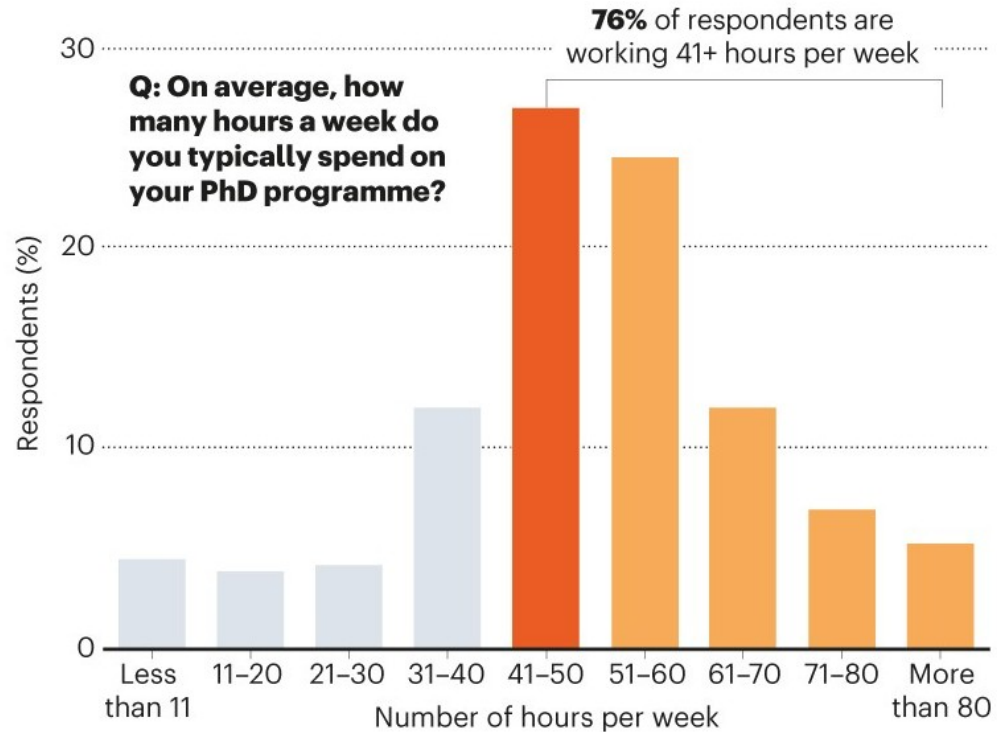
*From a 6000 PhD student survey
by nature. (Woolston 2019)*

Evidence 4: PhD students are overworked



OVEREXTENDED AND STRESSED

Long hours in the laboratory and other demands have taken a toll on PhD students' well-being and mental health.



From a 6000 PhD student survey by nature. (Woolston 2019)

Evidence 5: everybody is overwhelmed

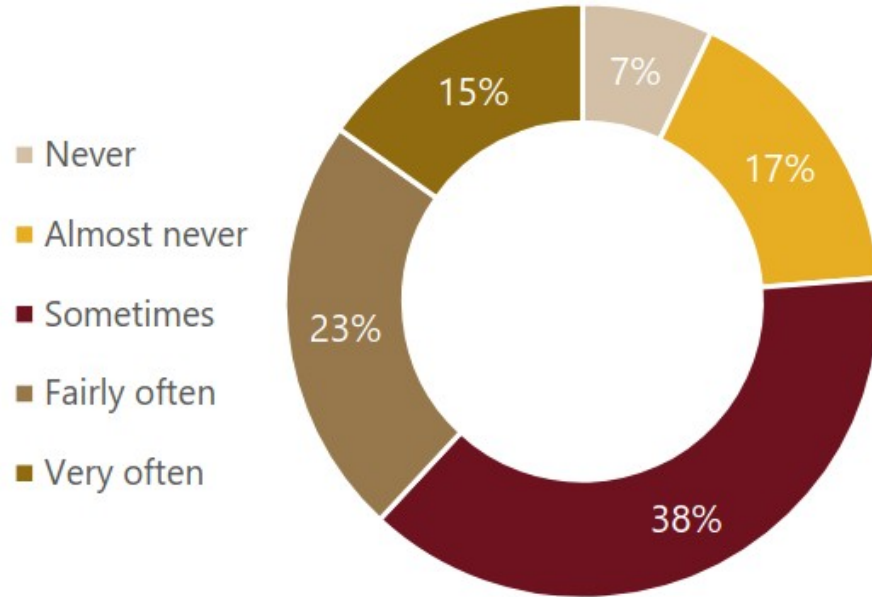
From a worldwide survey (Cerejo, Awati, and Hayward 2020)



Evidence 5: everybody is overwhelmed

From a worldwide survey (Cerejo, Awati, and Hayward 2020)

In the last month, how often have you felt overwhelmed by your situation at work?



38% stated that they had felt overwhelmed by their work situation fairly or very often in the previous month

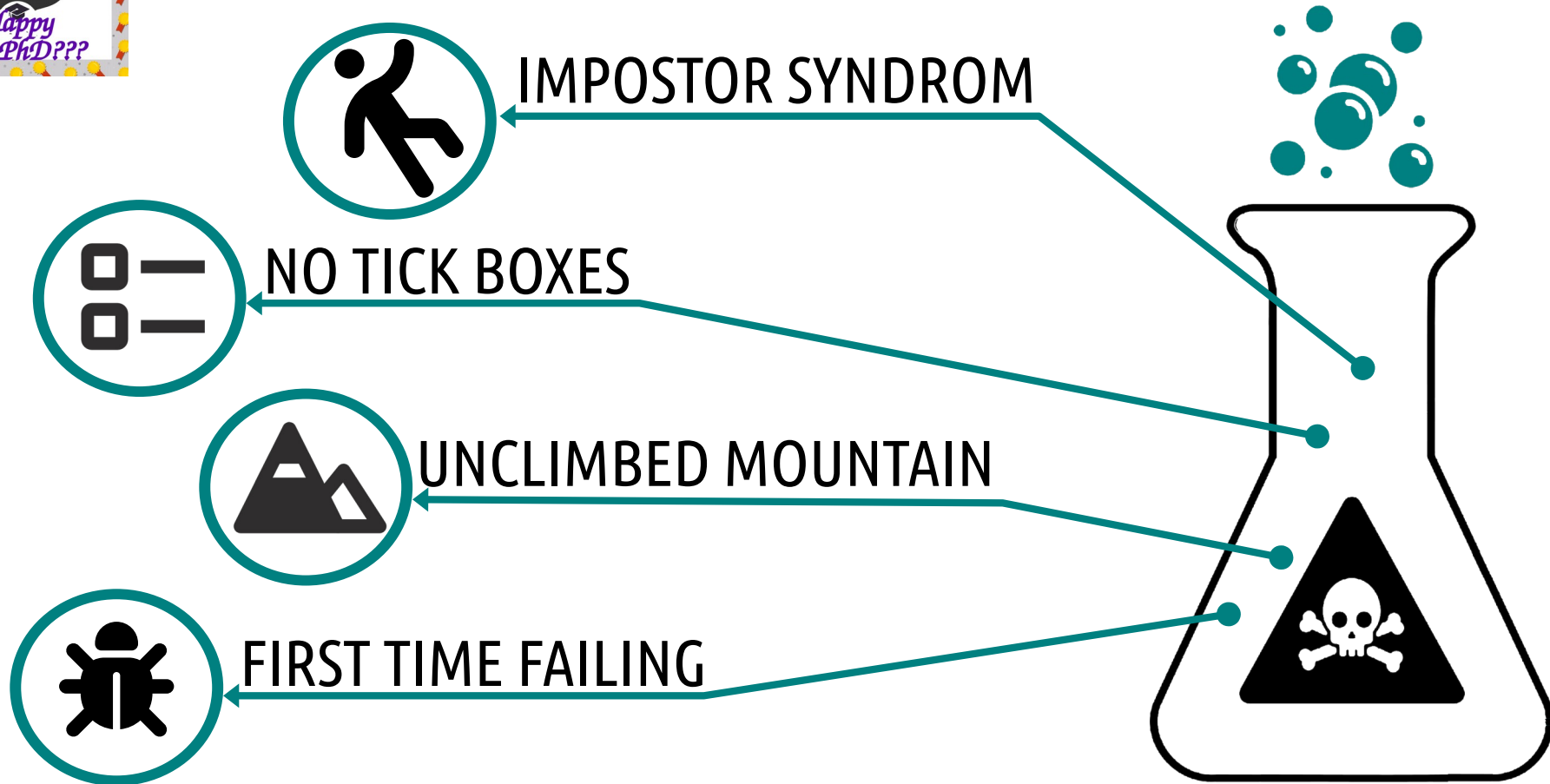
$n = 10,765$



Hypothesis 2: research is by essence hard



Hypothesis 2: research is by essence hard



Hypothesis 2: research is by essence hard



Ok, fair point, but...

Hypothesis 2: research is by essence hard



Ok, fair point, but...

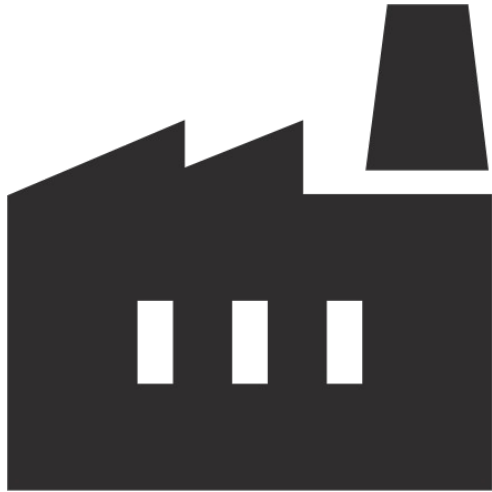
Harder than most
other jobs????

Hypothesis 2: research is by essence hard



Ok, fair point, but...

Harder than most
other jobs????



Unwell-being in academia?



The missing link

In all academia:

- too much unwell being
- a leaky pipeline

Starting to tie things up?



The missing link

In all academia:

- too much unwell being
- a leaky pipeline

Worse at the bottom
of the power ladder



The missing link

In all academia:

- too much unwell being
- a leaky pipeline

Worse at the bottom
of the power ladder

Less women the
higher we go in
the power ladder



The missing link

In all academia:

- too much unwell being
- a leaky pipeline

Worse at the bottom
of the power ladder

Less women the
higher we go in
the power ladder

Starting to tie things up?



We need
more facts!



Investigation - part 3

Evidence 6: strong inequalities in the access to a higher status



It is more likely*

Evidence 6: strong inequalities in the access to a higher status

2.4x

to get a PhD if a close
relative has one

It is

more likely*



Evidence 6: strong inequalities in the access to a higher status

2.4x

to get a PhD if a close
relative has one

It is

more likely*

2.6x

to become a
professor if male



Evidence 6: strong inequalities in the access to a higher status

2.4x

to get a PhD if a close
relative has one

It is

more likely*

2.6x

to become a
professor if male

4.6x

to become uni staff without
a declared disability

Omnipresent discriminations

*pure statistical single-variable analysis



Evidence 6: strong inequalities in the access to a higher status



2.4x

to get a PhD if a close
relative has one

It is **6.2x** more likely* to become a professor if
white-skinned instead of black

2.6x

to become a
professor if male

4.6x

to become uni staff without
a declared disability

Omnipresent discriminations

*pure statistical single-variable analysis

Evidence of inequalities in the access to a higher status

- + Weight-related discrimination
- + LGBTQIA+ related discrimination
- + Beliefs related discrimination
- + Money based discrimination
- +

4.6X

Omnipresent discriminations

to become uni staff without
a declared disability

*pure statistical single-variable analysis



It is



Evidence

It is

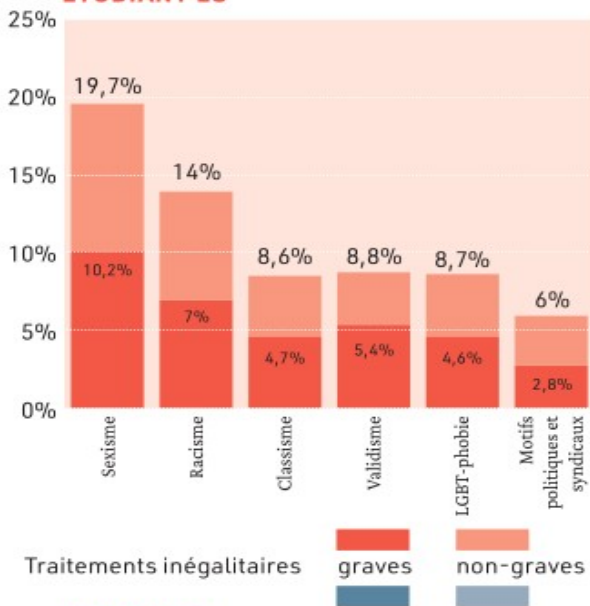
+ W
+ L
+ B
+ M
+ ..

4.6x

DES CHIFFRES / L'expérience de traitements inégaux par motif discriminatoire à l'université en France

POURCENTAGE DE PERSONNES DÉCLARANT AVOIR ÉTÉ VICTIME DE TRAITEMENTS INÉGALITAIRES PAR MOTIF DISCRIMINATOIRE

ÉTUDIANT-ES



Traitements inégaux

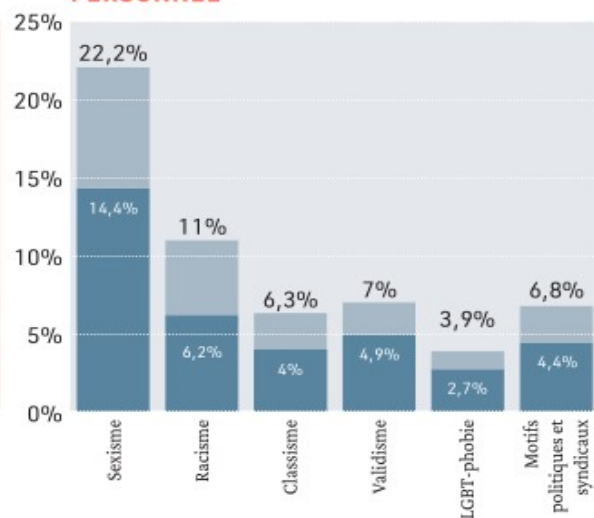


How to do good research?

in the access to a higher status

discrimination

PERSONNEL



to become uni staff without a declared disability

*pure statistical single-variable analysis

Omnipresent discriminations

Evidence 7: men are not competing fairly



Evidence 7: men are not competing fairly

- *Men spend more time doing research, and leave the less rewarding work to women. (Winslow 2010, O'Meara et al. 2017)*



Evidence 7: men are not competing fairly



- *Men spend more time doing research, and leave the less rewarding work to women. (Winslow 2010, O'Meara et al. 2017)*
- *Men self-cite 70 percent more than women. (King et al. 2017)*



Evidence 7: men are not competing fairly

- *Men spend more time doing research, and leave the less rewarding work to women. (Winslow 2010, O'Meara et al. 2017)*
- *Men self-cite 70 percent more than women. (King et al. 2017)*
- *Female authored papers are less cited (Bendels et al. 2018), and notably less likely to be cited by men. (Koffi 2021)*



Evidence 7: men are not competing fairly

- *Men spend more time doing research, and leave the less rewarding work to women. (Winslow 2010, O'Meara et al. 2017)*
- *Men self-cite 70 percent more than women. (King et al. 2017)*
- *Female authored papers are less cited (Bendels et al. 2018), and notably less likely to be cited by men. (Koffi 2021)*
- *Parenthood leads to a significant decline in the number of publications by women, while not for men. (Lutter and Schröder 2020)*

Evidence 8: evaluations are highly biased





Evidence 8: evaluations are highly biased

- *A women name on a CV implies evaluation bias.
(Steinpreis, Anders, and Ritzke 1999)*



Evidence 8: evaluations are highly biased

- *A women name on a CV implies evaluation bias.
(Steinpreis, Anders, and Ritzke 1999)*
- *The “two-body” problem is only used to refuse women not seen as movable.
(Rivera 2017)*



Evidence 8: evaluations are highly biased

- *A women name on a CV implies evaluation bias. (Steinpreis, Anders, and Ritzke 1999)*
- *The “two-body” problem is only used to refuse women not seen as movable. (Rivera 2017)*
- *Women don’t get credit for group collaborations, while men do. (Sarsons 2017)*



Evidence 8: evaluations are highly biased

- *A women name on a CV implies evaluation bias. (Steinpreis, Anders, and Ritzke 1999)*
- *The “two-body” problem is only used to refuse women not seen as movable. (Rivera 2017)*
- *Women don’t get credit for group collaborations, while men do. (Sarsons 2017)*
- *Under-represented minorities faculty are 44% less likely to receive unanimous votes from P&T committees. (Masters-Waage et al. 2024)*



Evidence 8: evaluations are highly biased

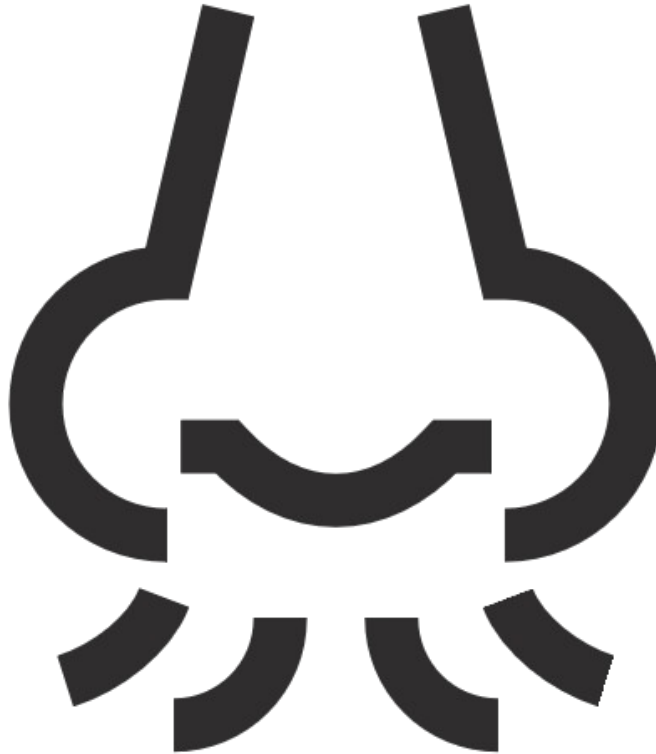
- *A women name on a CV implies evaluation bias. (Steinpreis, Anders, and Ritzke 1999)*
- *The “two-body” problem is only used to refuse women not seen as movable. (Rivera 2017)*
- *Women don’t get credit for group collaborations, while men do. (Sarsons 2017)*
- *Under-represented minorities faculty are 44% less likely to receive unanimous votes from P&T committees. (Masters-Waage et al. 2024)*
- *Career breaks are discriminated against in CVs. (Kristal et al. 2023)*



PAUSE !

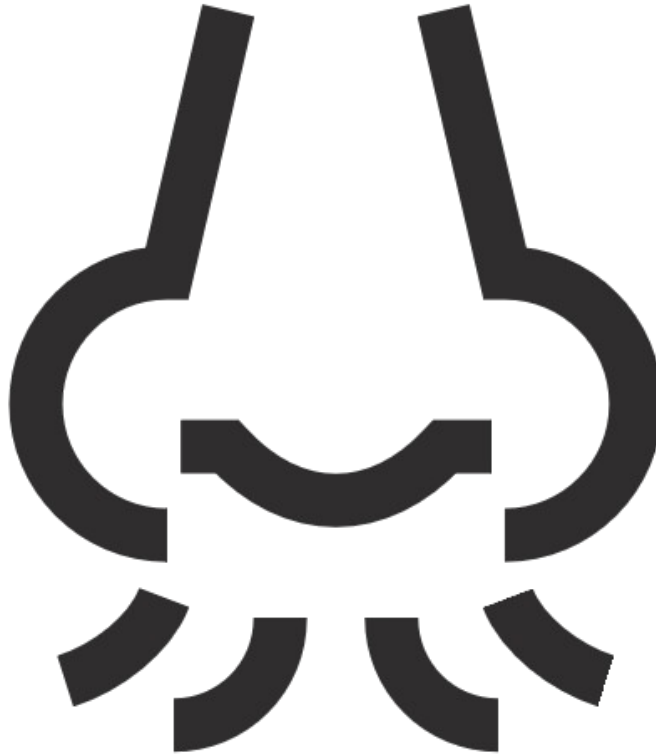


PAUSE !





PAUSE !



It is only getting worse from here :(

Evidence 9: men are **ACTIVELY** gate keeping

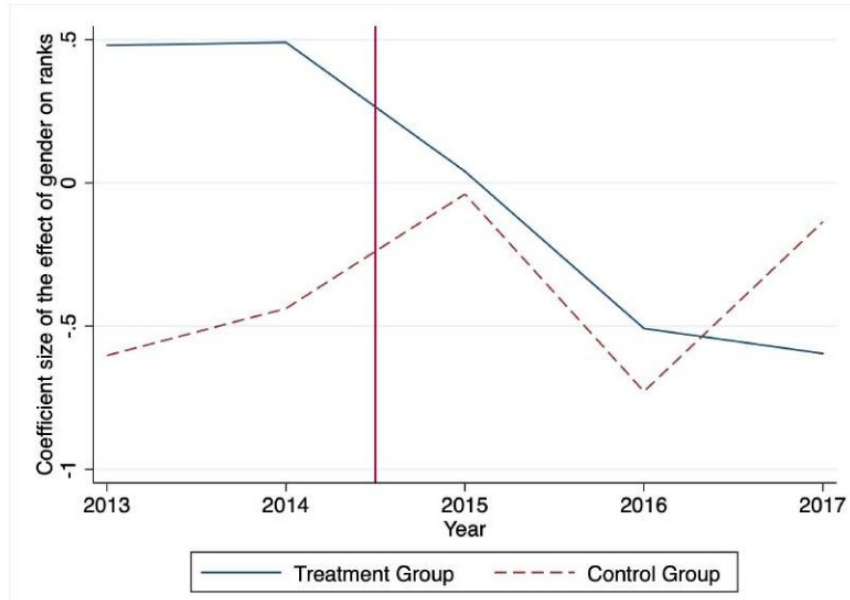
Consequences of the french 2015 reform « 40 % ratio » in committees on STEM fields



Evidence 9: men are **ACTIVELY** gate keeping

Consequences of the french 2015 reform « 40 % ratio » in committees on STEM fields

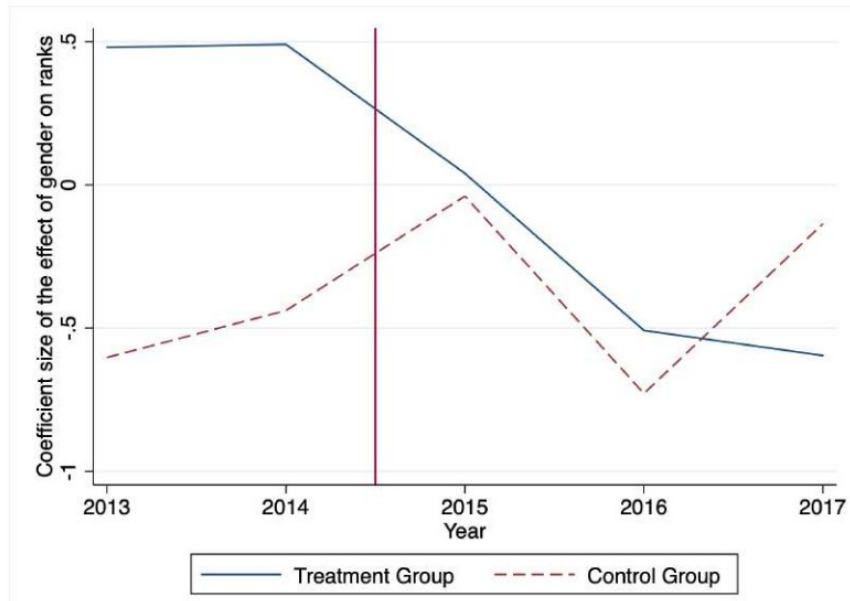
Figure 3: The effect of gender on ranks from a rank-ordered logit, controlling for publications



Evidence 9: men are **ACTIVELY** gate keeping

Consequences of the french 2015 reform « 40 % ratio » in committees on STEM fields

Figure 3: The effect of gender on ranks from a rank-ordered logit, controlling for publications



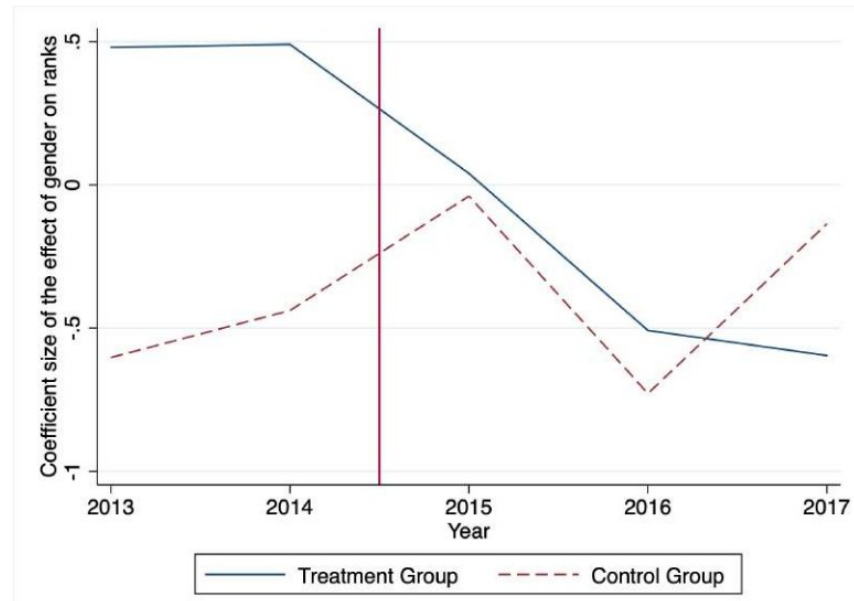
« the negative effect of the reform is concentrated in committees headed by men, this result seems driven by the reaction of men to the reform »

(Deschamps 2024)

Evidence 9: men are **ACTIVELY** gate keeping

Consequences of the french 2015 reform « 40 % ratio » in committees on STEM fields

Figure 3: The effect of gender on ranks from a rank-ordered logit, controlling for publications



« the negative effect of the reform is concentrated in committees headed by men, this result seems driven by the reaction of men to the reform »

(Deschamps 2024)

« male evaluators become less favorable toward female candidates as soon as a female evaluator joins the committee »

(Bagues, Sylos-Labini, and Zinovyeva 2017)



Fun (?) Fact 1:

***you are more likely to disbelieve
facts demonstrating biases if you are a man!***
(Handley et al. 2015)



Fun (?) Fact 1:

***you are more likely to disbelieve
facts demonstrating biases if you are a man!***
(Handley et al. 2015)

In fact, it is often that our brains don't see some things (remember the previous illusion), and it can be very hard to finally see them. But, we need to see more things to unlock the final piece of the puzzle.

Omnipresent discriminations



Investigation - part 4



How to do good research?

The magical hiding glasses ? (by Privileged IncTM)



The final piece of the puzzle



How to do good research?

The magical hiding glasses ? (by Privileged Inc™)



The final piece of the puzzle



How to do good research?

The magical hiding glasses ? (by Privileged Inc™)

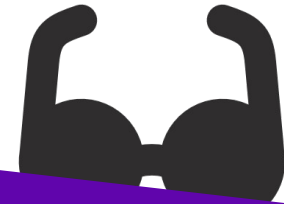


The final piece of the puzzle

How to do good research?



The most privileged Inc™)



If we are not the target of a
discriminating or harassing behaviour
we will probably **not even see it**

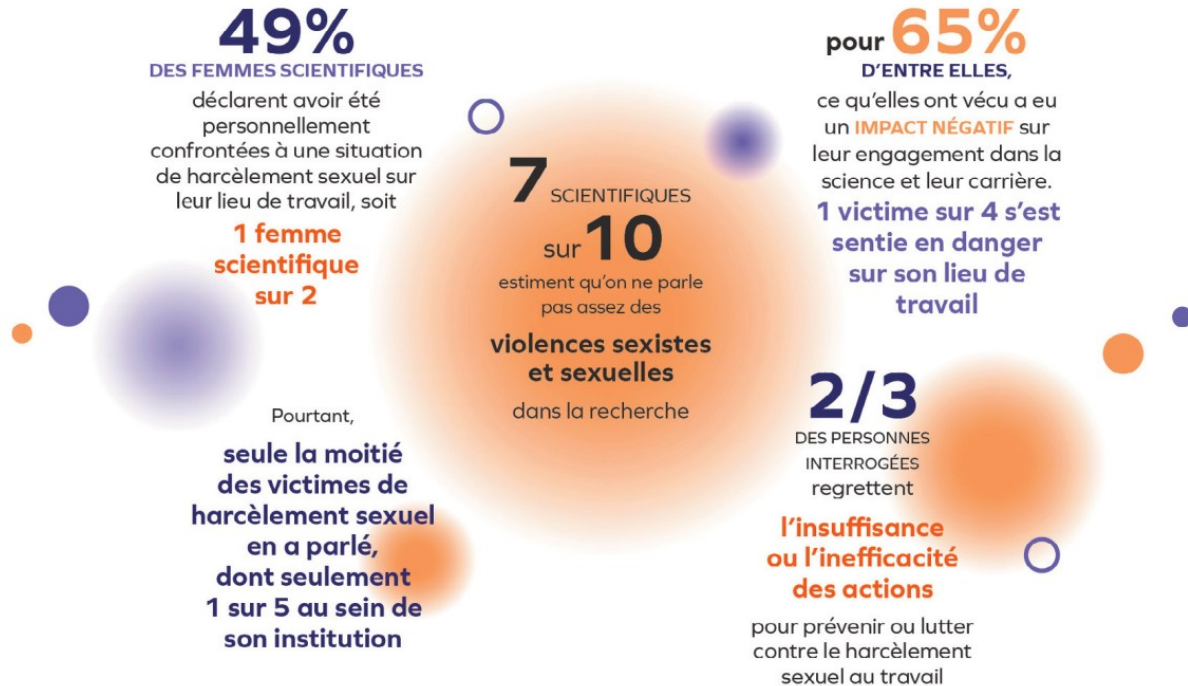


The final piece of the puzzle

Evidence 10: the omnipresent « invisible » phenomenon



Evidence 10: the omnipresent « invisible » phenomenon



Evidence 10: the omnipresent « invisible » phenomenon

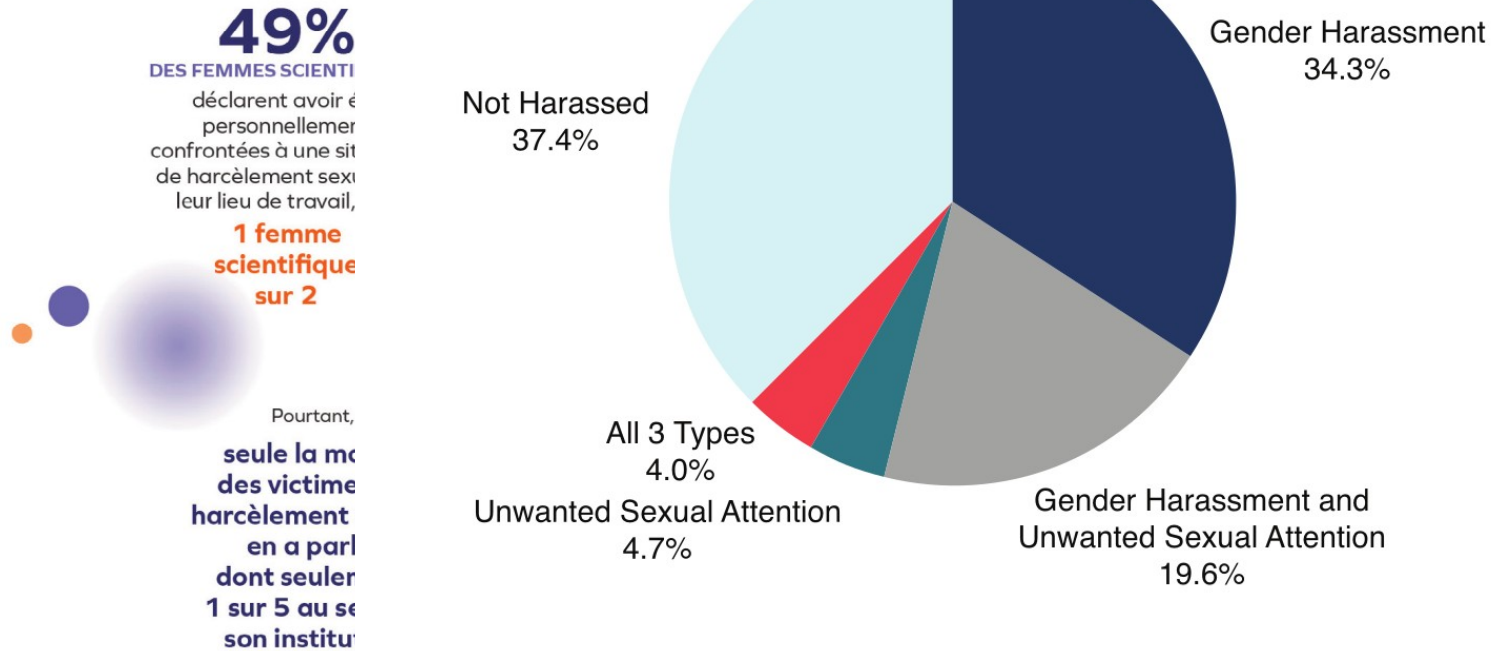
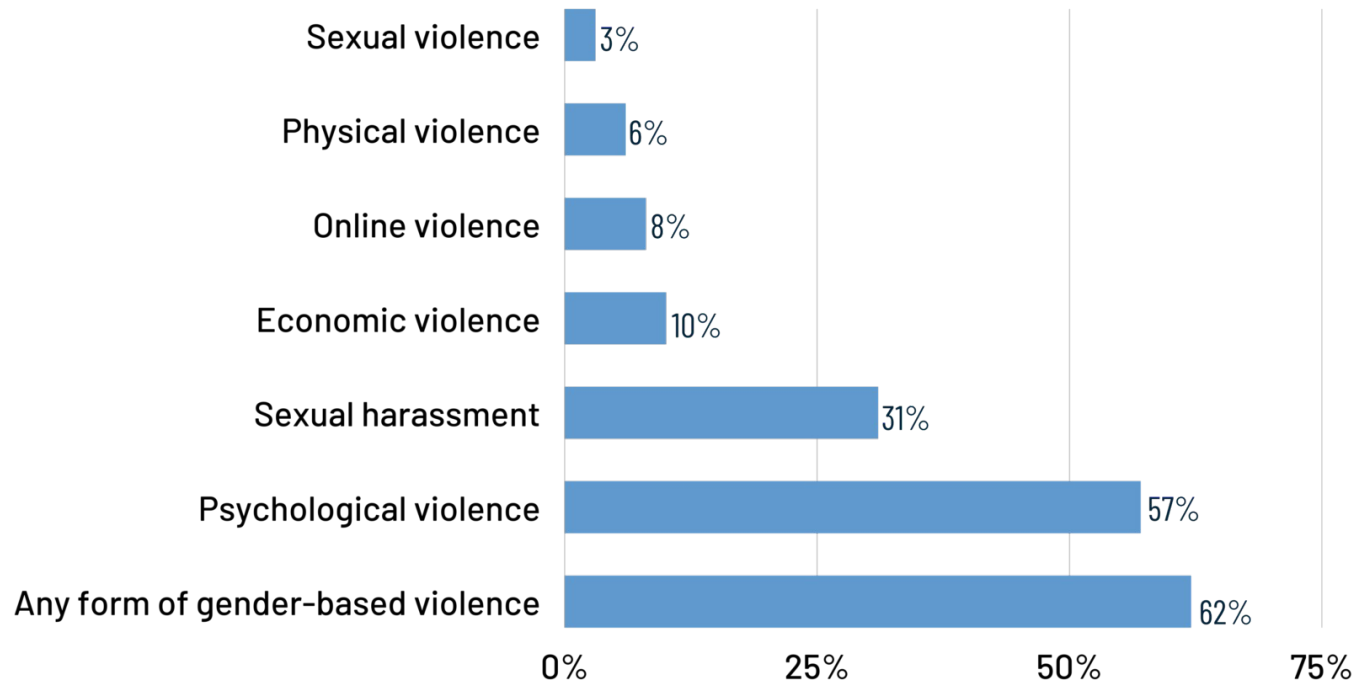


FIGURE 3-1 Percentage of types of sexual harassment experiences among female university employees.



Prevalence of any form of gender-based violence and by form of gender-based violence

Source of data: Lipinsky, Anke; Schredl, Claudia; Baumann, Horst; Humbert, Anne Laure; Tanwar, Jagriti; Bondestam, Fredrik; Freund, Frederike; Lomazzi, Vera (2022). UniSAFE Survey – Gender-based violence and institutional responses. GESIS – Leibniz Institut für Sozialwissenschaften. Data file Version 1.0.0, <https://doi.org/10.7802/2475>

ii-

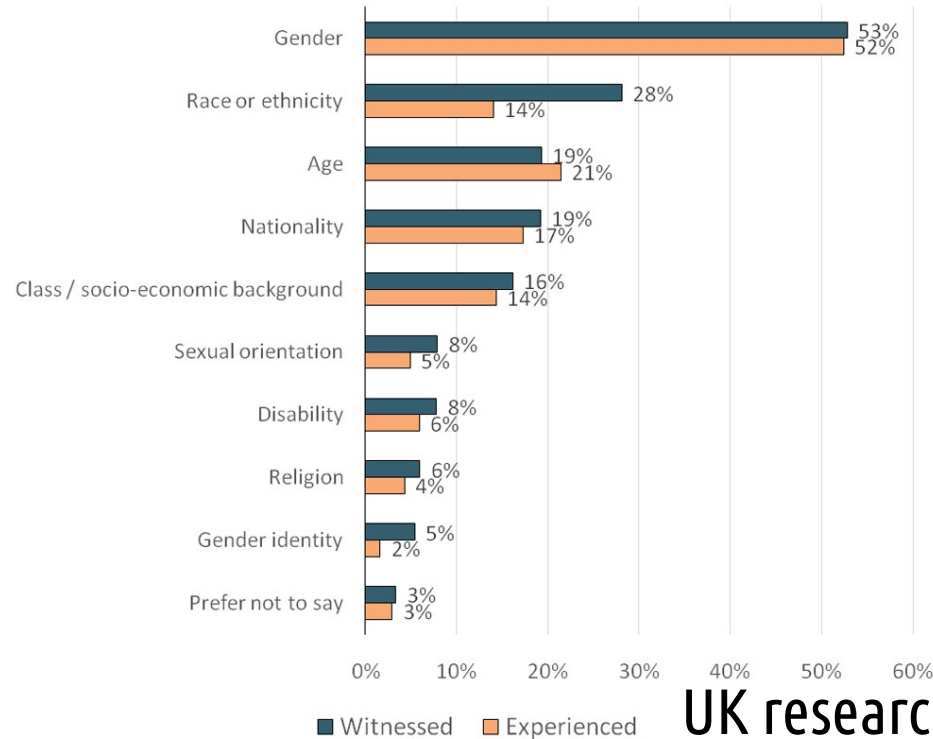
Evidence 11: it targets many groups at all levels



Evidence 11: it targets many groups at all levels



Q: In cases where you have witnessed or experienced bullying and harassment or discrimination, was this behaviour related to...



UK researchers

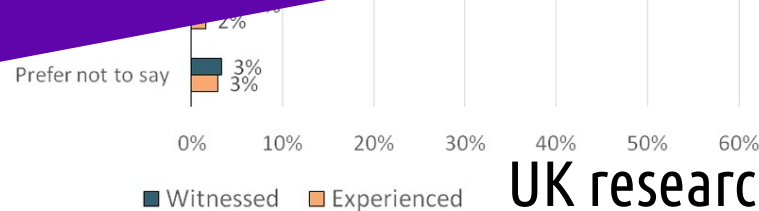
Evidence 11: it targets many groups at all levels



Q: In cases where you have witnessed or experienced bullying and harassment, how does this behaviour related to...

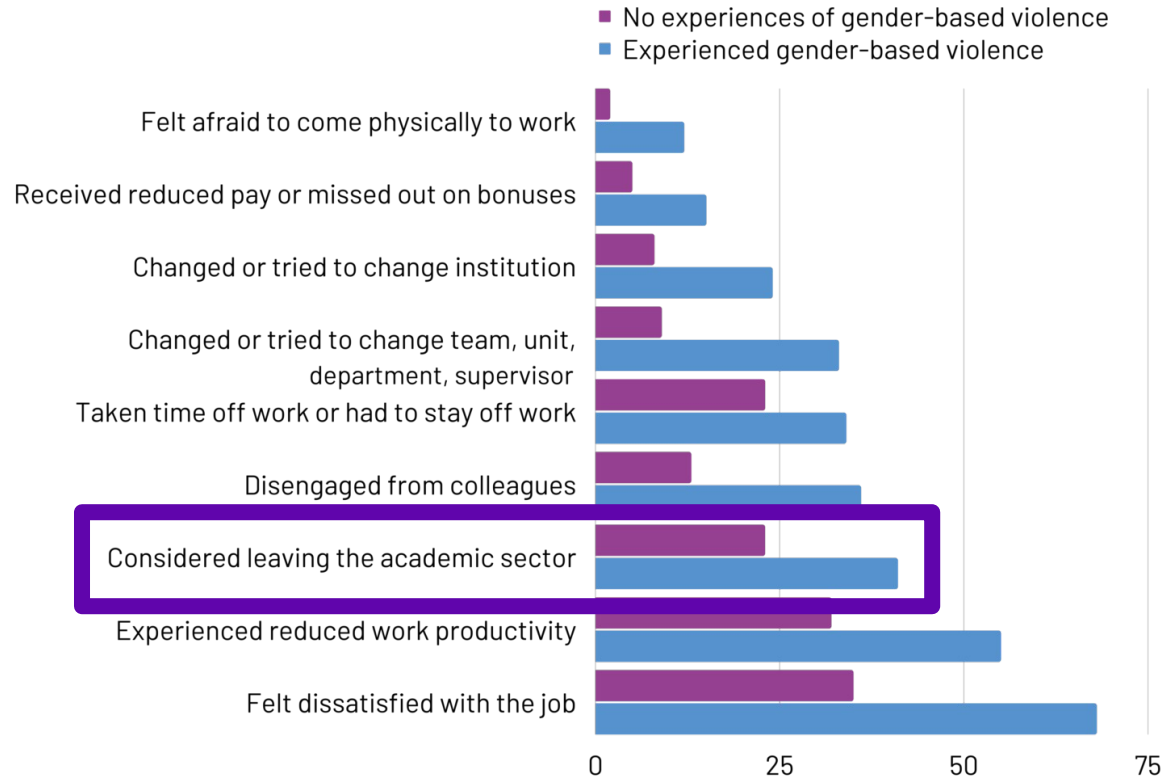
Since their arrivals in french higher education, the share of students target of a sexual assault, often reiterated, is:

- 24% of women
- 9% of men
- 33% of transgenders/non binaries/queers



UK researchers

Evidence 12: it has consequences



Source of data: Lipinsky, Anke; Schredl, Claudia; Baumann, Horst; Humbert, Anne Laure; Tanwar, Jagriti; Bondestam, Fredrik; Freund, Frederike; Lomazzi, Vera (2022). UniSAFE Survey – Gender-based violence and institutional responses. GESIS – Leibniz Institut für Sozialwissenschaften. Datenfile Version 1.0.0, <https://doi.org/10.7802/2475>



The missing link (part 2)

**Harassment and bullying is widespread
against women and under-represented minorities**

- It actively makes them run away
- It actively depresses them

The final piece of the puzzle



The missing link (part 2)

Harassment and bullying is widespread against women and under-represented minorities

- It actively makes them run away
- It actively depresses them

Worse at the bottom
of the power ladder



The missing link

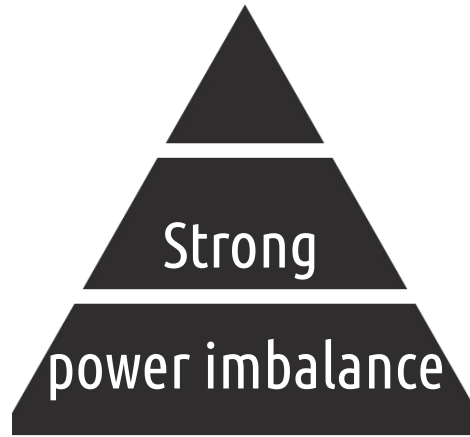
*Power imbalances foster and sustain sexual-harassment
(Benya, Widnall, and Johnson 2018) (Zara et al. 2024) (Bergeron et al. 2025),
and in fact other form of harassments and discriminations.
(Dhume and Roberti-Lintermans 2024)*



The missing link

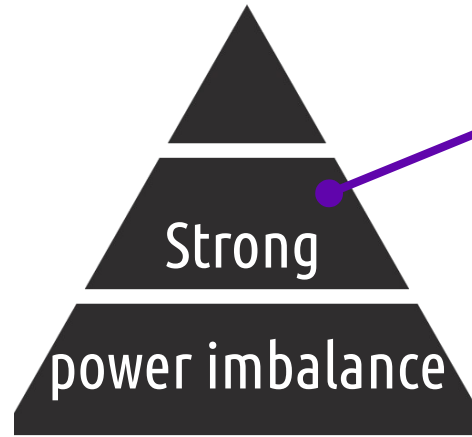
*Power imbalances foster and sustain sexual-harassment
(Benya, Widnall, and Johnson 2018) (Zara et al. 2024) (Bergeron et al. 2025),
and in fact other form of harassments and discriminations.
(Dhume and Roberti-Lintermans 2024)*

*« The academic workplace has the **second highest rate of sexual harassment** at 58 percent (the military has the highest rate at 69 percent) when comparing it with military, private sector, and the government »(Benya, Widnall, and Johnson 2018)*





How to do good research?



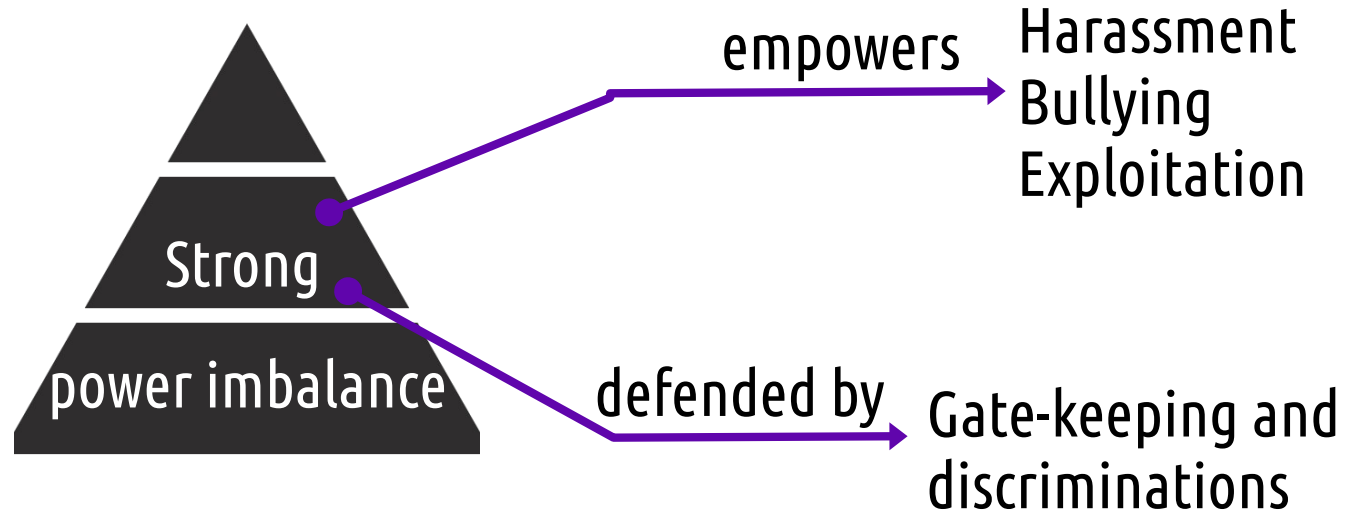
empowers

Harassment
Bullying
Exploitation

The final piece of the puzzle



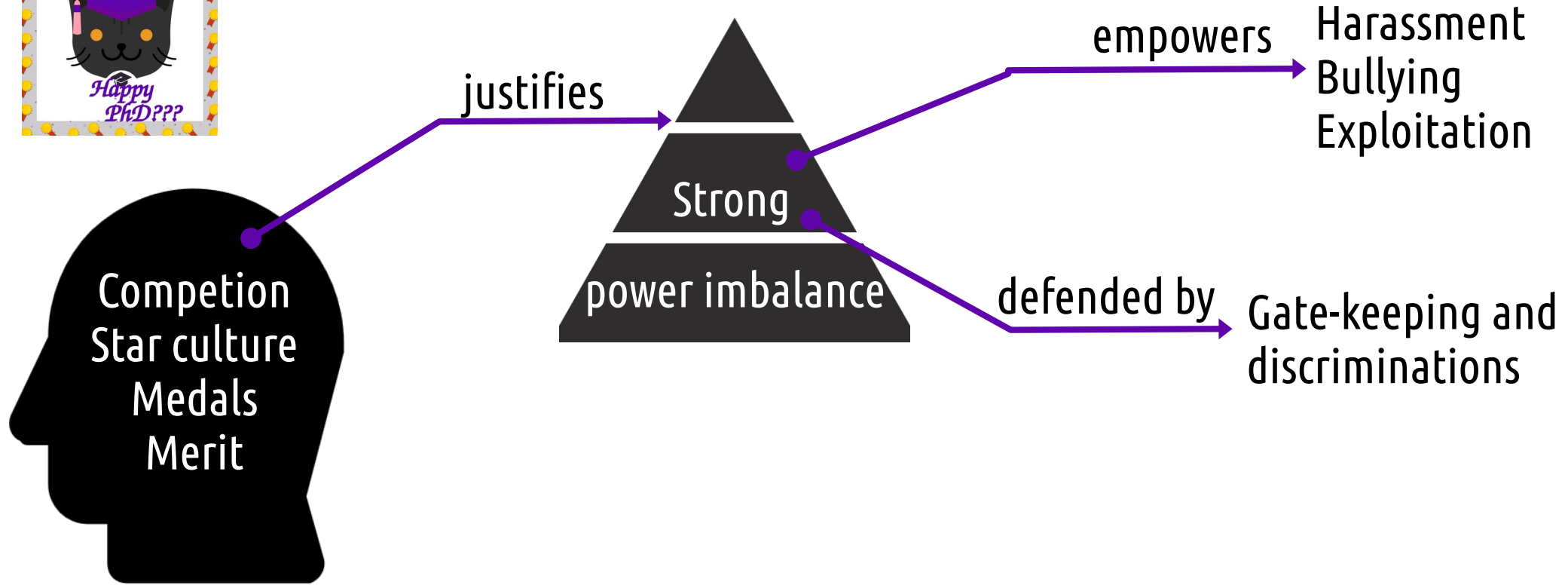
How to do good research?



The final piece of the puzzle



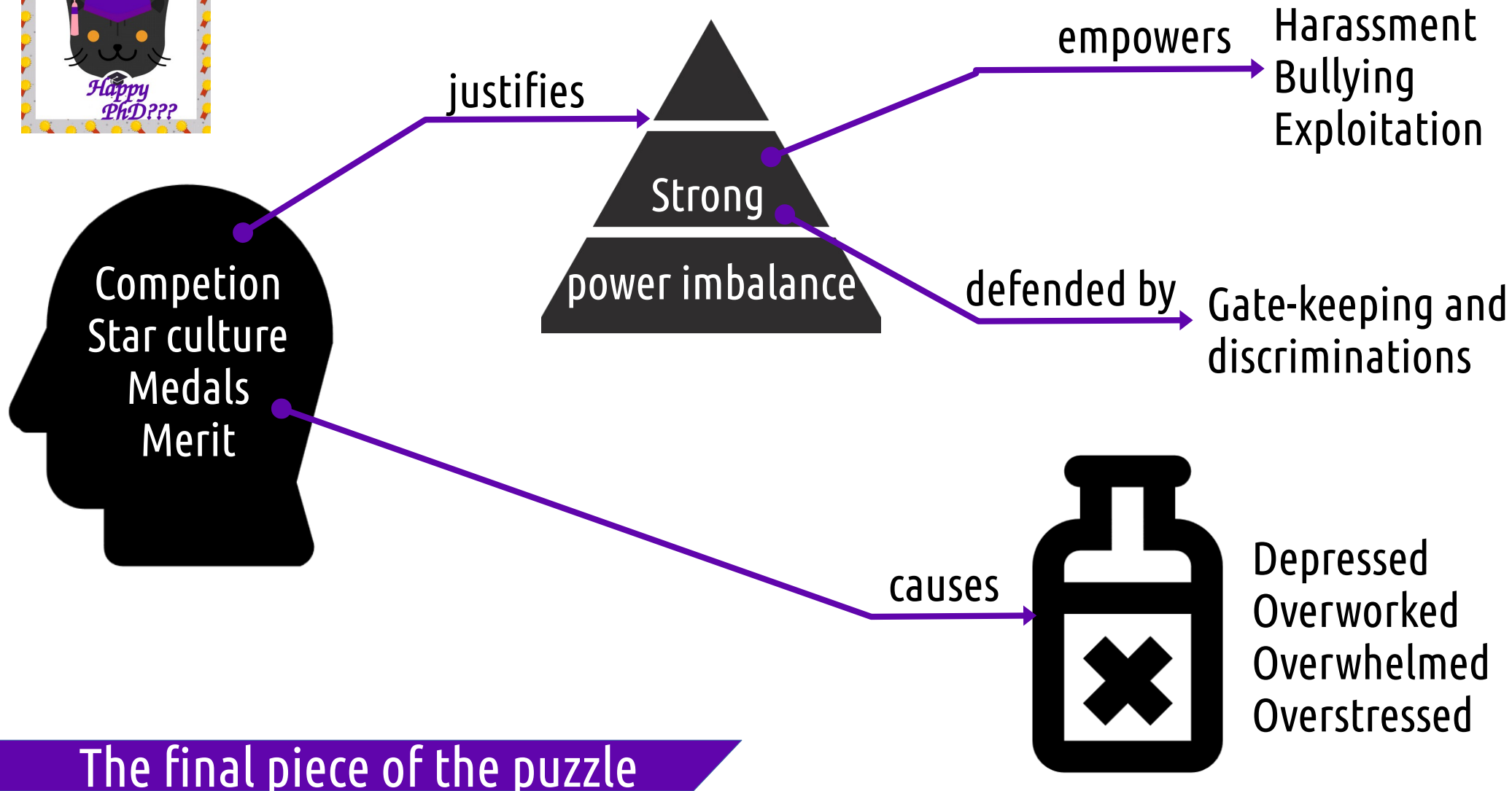
How to do good research?



The final piece of the puzzle

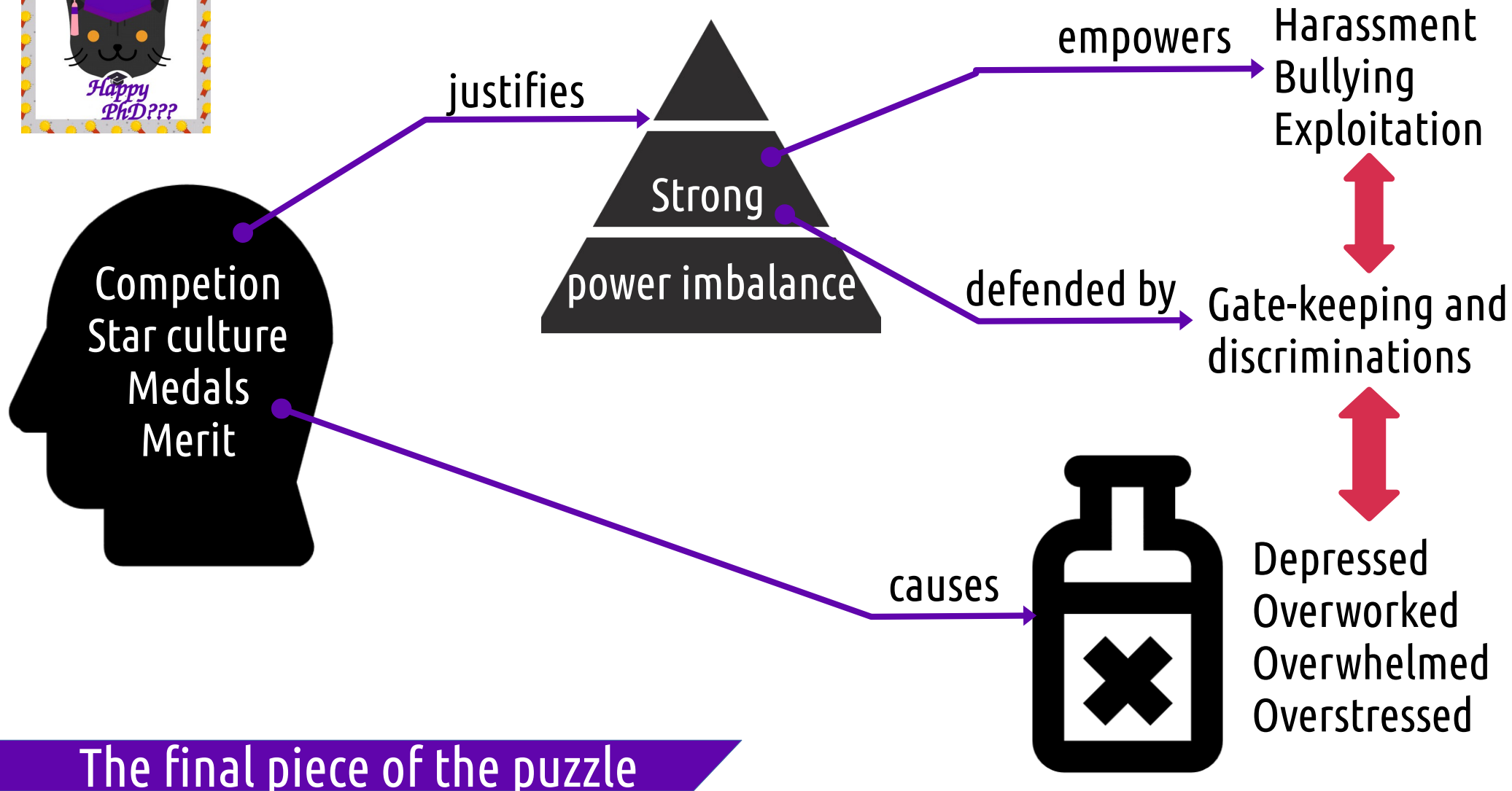


How to do good research?





How to do good research?





Ok, all this sounds bad...

Is it an issue?



Ok, all this sounds bad...
but it is only individual consequences,
and maybe « SCIENCE » and « PROGRESS » needs all this!



Ok, all this sounds bad...
but it is only individual consequences,
and maybe « SCIENCE » and « PROGRESS » needs all this!

First off, I find you morally diubious if you believe this, but ok, let's still be thorough.

Is it an issue?



Moral answer:

It is widely accepted that science with experiments that involve **misstreating human beings** is **unethical** and should not be done.



Moral answer:

It is widely accepted that science ~~with experiments~~ that involve **misstreating human beings** is **unethical** and should not be done.



Practical answer:

A lack of diversity leads to biased and dangerous science:

- airbags systems have been designed historically for men body shapes, leading to **less chances for women to survive car crashes**
- **medicine is dosed for men**, with few women in cohorts (19% for antiviral, 38% for vaccins, 11% for aids)
- women have distinct symptoms sometimes – e.g. yentl syndrom for cardiovascular diseases, **women with such conditions have 10% more chances of dying** when arriving at the hospital.
- **LLM with racist or sexist bias**, now deployed in companies to filter CVs.



An answer accepting the bad « progress » premise:

Diversity breeds innovation!

- publications by mixed-gender teams are more cited (Campbell et al. 2013)
- science by mixed-gender teams is significantly more novel and impactful (Yang et al. 2022)
- diversity in inputs by author ethnicity, location, and references leads to greater contributions to science as measured by impact factors and citations. (Freeman and Huang 2015)
- diverse scholars are more innovative, but their work is taken up by others less than it should be (Hofstra et al. 2020)

Is it an issue?



We have many issues and we understand them and their roots.



We have many issues and we understand them and their roots.

TIME TO ACT

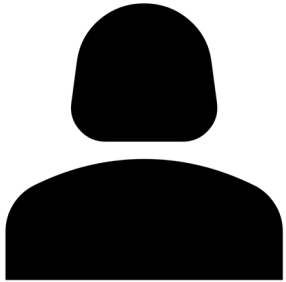
What exists and what beliefs are behind it





What exists and what beliefs are behind it

Individual

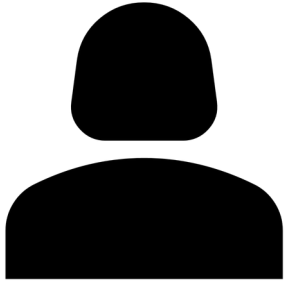


Mentoring
Networking



What exists and what beliefs are behind it

Individual



Mentoring
Networking

Societal

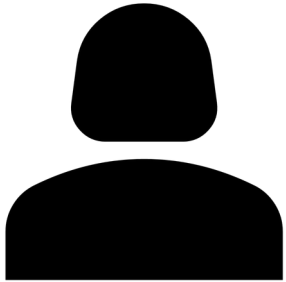


Child-care
Part-time
Bias training



What exists and what beliefs are behind it

Individual



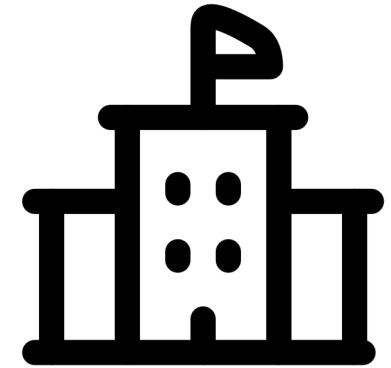
Mentoring
Networking

Societal



Child-care
Part-time
Bias training

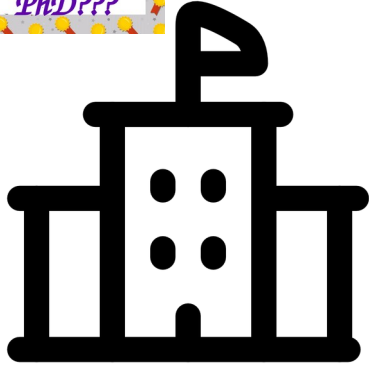
Institutional/Cultural



Fight harassment
Train employees
Improve the culture



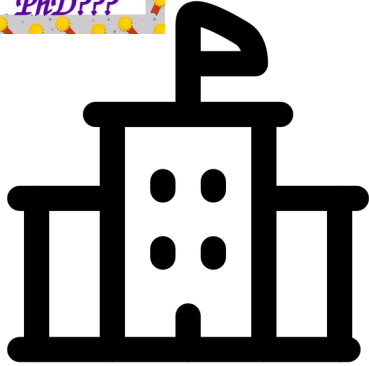
Are institutions doing enough ?



often focus on solutions from the individual or societal point of view, it is either women who are missing something or society to blame, but never our academic culture which is blamed.



Are institutions doing enough ?

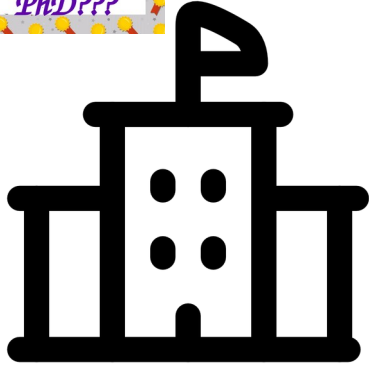


often focus on solutions from the individual or societal point of view, it is either women who are missing something or society to blame, but never our academic culture which is blamed.

They **do not address** the most common form of harassment (**gender-harassment**), provide **opaque and inefficient report procedures** (He et al. 2024) leading to a feeling of betrayal (Bergeron et al. 2025).



Are institutions doing enough ?



often focus on solutions from the individual or societal point of view, it is either women who are missing something or society to blame, but never our academic culture which is blamed.

They **do not address** the most common form of harassment (**gender-harassment**), provide **opaque and inefficient report procedures** (He et al. 2024) leading to a feeling of betrayal (Bergeron et al. 2025).

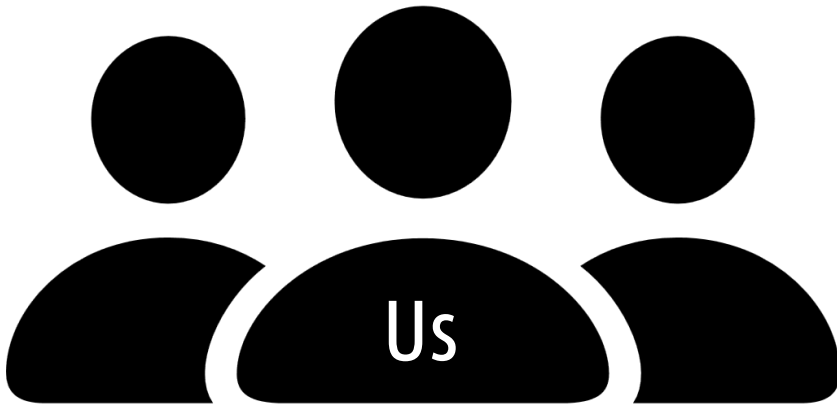
They often focus on **protecting the image of the institution** (Faniko et al. 2021), and on just aiming for **legal compliance**, which had little impact on harassment, (Benya, Widnall, and Johnson 2018), notably as most targets of harassment don't report issues. (Lipinsky et al. 2022)

A fourth point of view ?





A fourth point of view ?

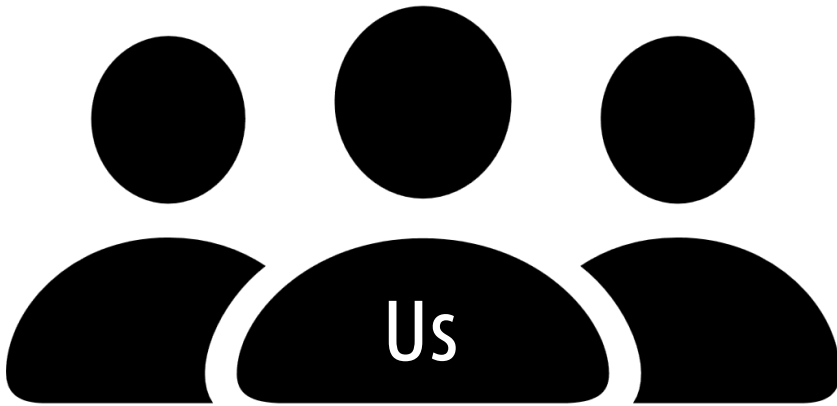


How to do good research?

A fourth point of view ?



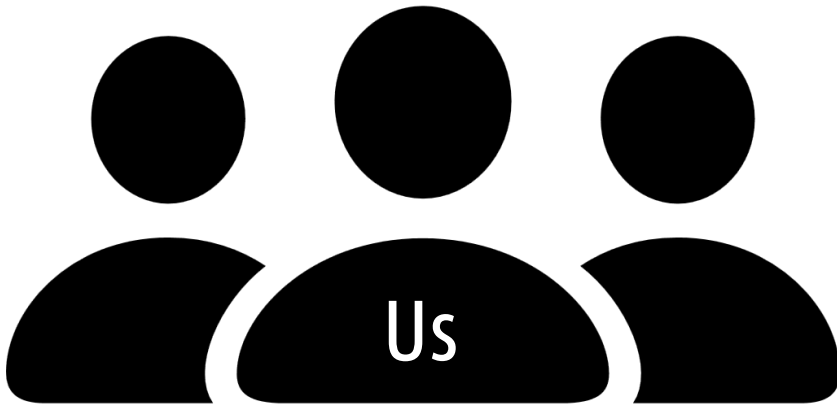
**We do the research, we are part
of this toxic culture, and we have
power over it.**



Act



A fourth point of view ?

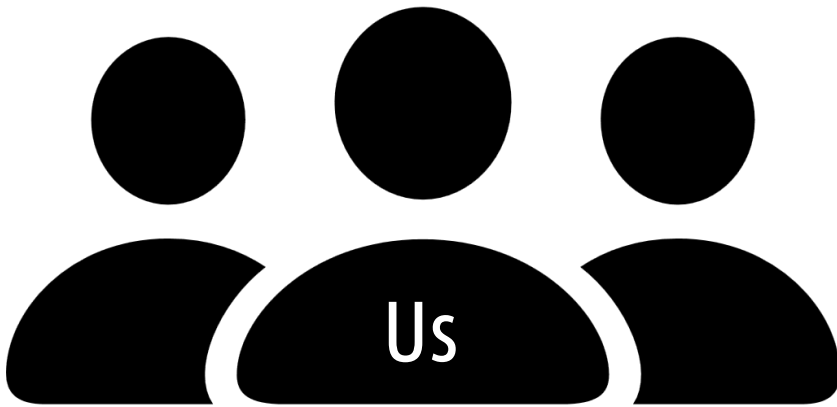


We do the research, we are part of this toxic culture, and we have power over it.

« A systemwide change to the culture and climate in higher education is required to prevent and effectively address all three forms of sexual harassment. » (

Benya, Widnall, and Johnson 2018)

A fourth point of view ?



We do the research, we are part of this toxic culture, and we have power over it.

« A systemwide change to the culture and climate in higher education is required to prevent and effectively address all three forms of sexual harassment. » (Benya, Widnall, and Johnson 2018)

« We need to focus on broad cultural change » (Johnson and Hoover 2015)



Step 1: be the bare minimal of reasonable human beings

The basics, a few general rules:

- Don't touch or be very close to people at the workplace
- Don't comment on the physical appearance of people
- Avoid one on one meetings with subordinates outside of the workplace
- Don't try to date your students
- Don't ever say somebody does not belong here because of X or Y
- Don't make general comments about some group X having some supposed characteristics
- ...

How to do good research?

Step 2: question the toxic culture, spread a better mentality



42% of PhD students think it's normal to suffer mentally.

50% don't feel they can talk about not pursuing academia



How to do good research?

Step 2: question the toxic culture, spread a better mentality

Relieve the pressure:

- It's normal to not work outside of the office.
- It's normal if research is not my whole life.
- It's ok if I don't publish one more paper this year.
- It's ok if I go on long vacations.
- It's ok if I don't want to pursue research after my PhD.
- It's ok if I am not doing a permanent crunch.
- It's especially important not to be « brilliant ».

42% of PhD students think it's normal to suffer mentally.

50% don't feel they can talk about not pursuing academia

Step 3: don't be « brillant », don't create superstars, research is a collective



Merit is



Step 3: don't be « brillant », don't create superstars, research is a collective

Merit is

- not the main factor of academic success

Step 3: don't be « brilliant », don't create superstars, research is a collective



Merit is

- not the main factor of academic success
- often missattributed
(unvoluntarily, but also voluntarily...)

Step 3: don't be « brilliant », don't create superstars, research is a collective



Merit is

- not the main factor of academic success
- often missattributed
(unvoluntarily, but also voluntarily...)

Bellman-Ford
Cantor set
Chernoff Bound
Currying
De Morgan's law
Floyd-Warshall
Gauss' theorem
Gaussian elimination
Grobner basis
Moore's law
Russel paradox
Von Neuman Archi

Step 3: don't be « brillant », don't create superstars, research is a collective



Merit is

- not the main factor of academic success
- often missattributed
(unvoluntarily, but also voluntarily...)
- a belief which triggers resistance against egalitarian action (**Faniko et al. 2021**)

Bellman-Ford
Cantor set
Chernoff Bound
Currying
De Morgan's law
Floyd-Warshall
Gauss' theorem
Gaussian elimination
Grobner basis
Moore's law
Russel paradox
Von Neuman Archi

Step 3: don't be « brillant », don't create superstars, research is a collective



Merit is

- not the main factor of academic success
- often missattributed
(unvoluntarily, but also voluntarily...)
- a belief which triggers resistance against egalitarian action (Faniko et al. 2021)
- protecting harassers: the behavior of male colleagues perceived as “superstars” is often minimized or ignored (Benya, Widnall, and Johnson 2018)

Bellman-Ford
Cantor set
Chernoff Bound
Currying
De Morgan's law
Floyd-Warshall
Gauss' theorem
Gaussian elimination
Grobner basis
Moore's law
Russel paradox
Von Neuman Archi

Step 3: don't be « brillant », don't create superstars, research is a collective



Merit is

- not the main factor of academic success
- often missattributed
(unvoluntarily, but also voluntarily...)
- a belief which triggers resistance against egalitarian action (Faniko et al. 2021)
- protecting harassers: the behavior of male colleagues perceived as “superstars” is often minimized or ignored (Benya, Widnall, and Johnson 2018)
- a belief which discriminates against women in our society

Act

Bellman-Ford
Cantor set
Chernoff Bound
Currying
De Morgan's law
Floyd-Warshall
Gauss' theorem
Gaussian elimination
Grobner basis
Moore's law
Russel paradox
Von Neuman Archi



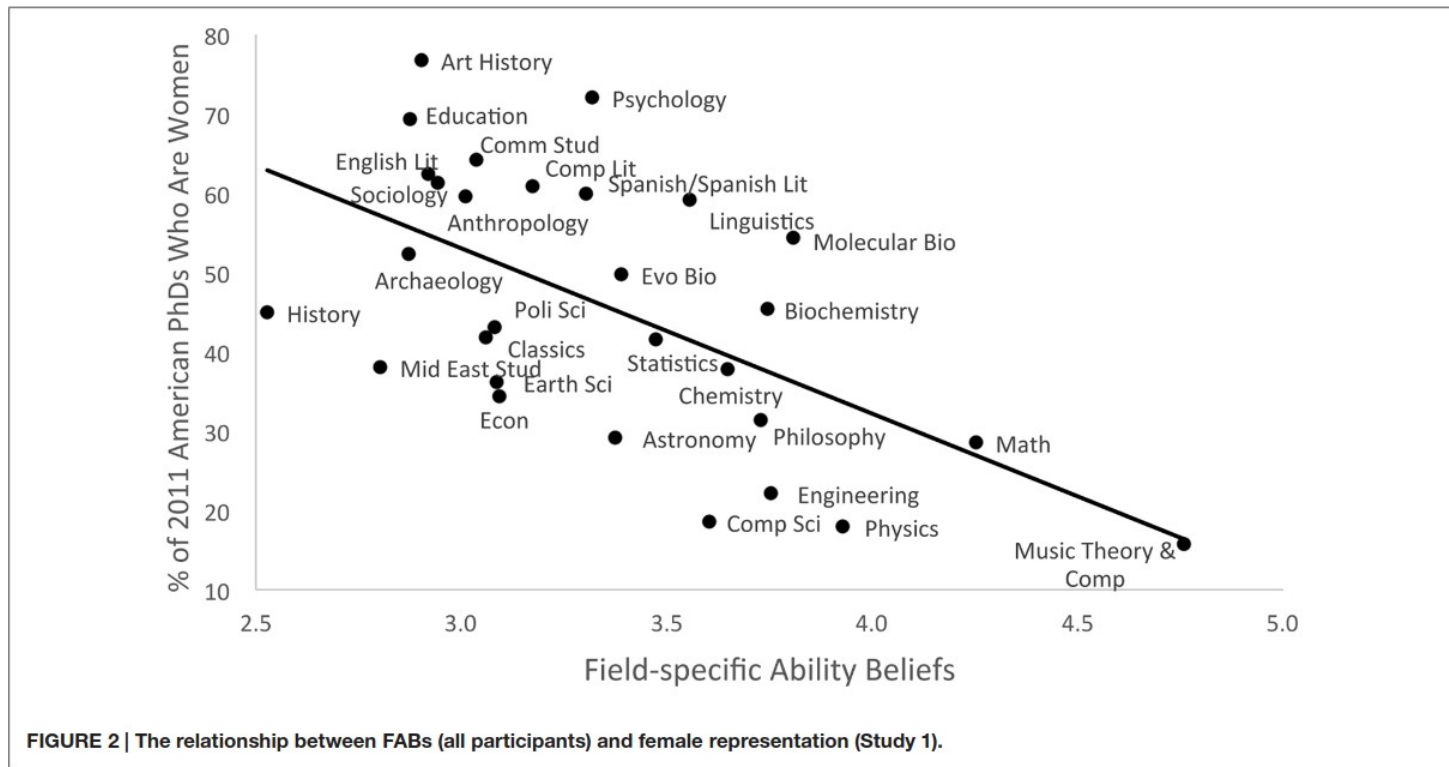
How to do good research?

Step 3: don't be « brilliant », don't create superstars, research is a collective

Merit is

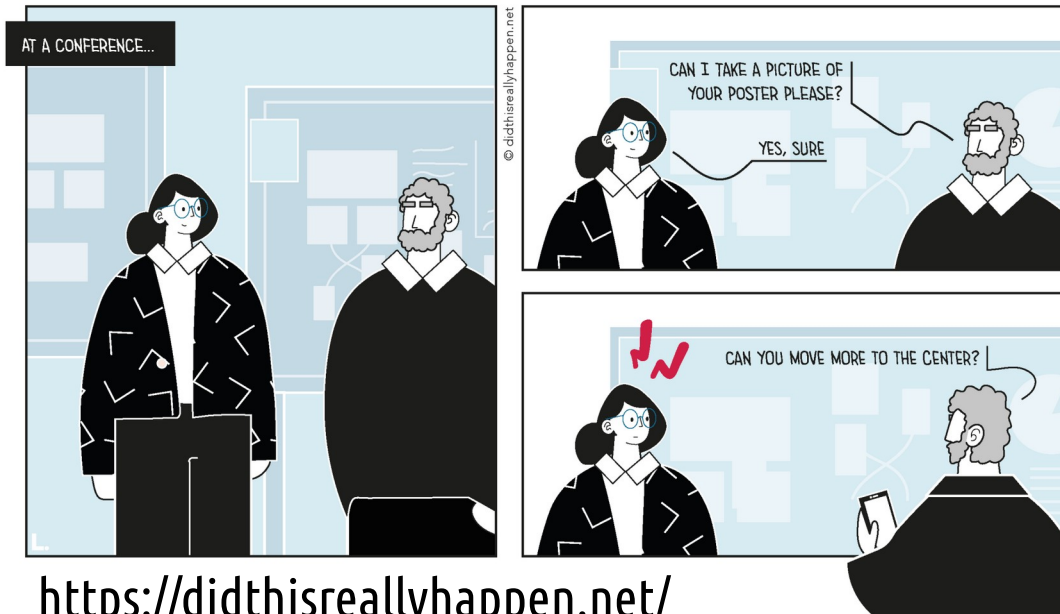
- not
- ofte
- (un
- a be
- acti
- prol
- perc
- (Be
- a be

Act



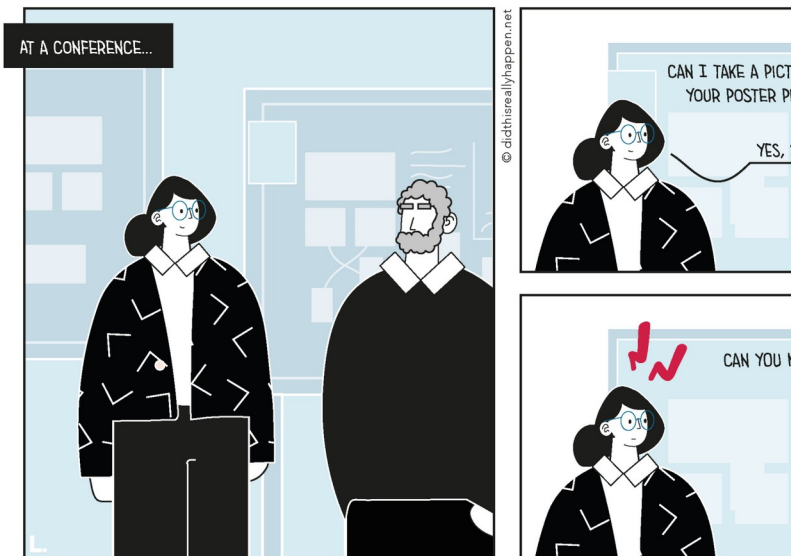
Bellman-Ford
Cantor set
Bernoff Bound
Currying
Morgan's law
loyd-Warshall
Gauss' theorem
Gaussian elimination
Grobner basis
Moore's law
Russel paradox
Von Neuman Archi

Step 4: recognize what is not acceptable





Step 4: recognize what



<https://didthisreallyhappen.net/>

Act

FIGHT AGAINST GENDER VIOLENCE IN ACADEMIA

The violence meter



o good research?



TO KNOW MORE

Step 5: don't be a passive bystander





Step 5: don't be a

FIVE

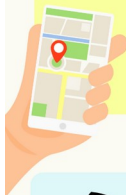
WAYS TO BE AN

ACTIVE BYSTANDER

2 DISTRACT

Interrupt by asking a question or starting a conversation. Try asking for directions or the time.

Best used when: You're unsure what the reaction might be to intervening directly.



4 DELAY

Check they're okay or if they need any help afterwards.

Best used when: You feel uncomfortable intervening at the time.

Are You Okay ?



Only intervene if it's safe.

1 DIRECT

Call the person out for their behaviour and let them know it's not okay.

Best used when: Among friends or people you know.

"LEAVE THEM ALONE"

3 DELEGATE

Ask someone else to help. This could be someone with authority or someone nearby.

Best used when: You need extra support.



5 DISAPPROVE

Send a clear signal you don't like what they're saying or doing. Try shaking your head or walking away.

Best used when: Among friends or people you know.



STAND
& STEP IN

5Ds taken from Brook's Active Bystander Course for universities.
Find out more at brook.org.uk/bystander-training/

brook

Act

to good research?

Step 7: learn to break the myths



Step 7: learn to break the myths

« There is risk of false accusations »

2 to 10 % at the very most
vs
60 % of gender-harassment?

Step 7: learn to break the myths



« There is risk of false accusations »

2 to 10 % at the very most
vs
60 % of gender-harassment?

« We are discriminating white men »

No, we are making things fair!
(did the french revolution
« discriminate » the nobles?)

Step 7: learn to break the myths



« There is risk of false accusations »

2 to 10 % at the very most
vs
60 % of gender-harassment?

« We are discriminating white men »

No, we are making things fair!
(did the french revolution
« discriminate » the nobles?)

« We can't say anything anymore »

It is the converse.
Most of « we » were never
able to say anything,
and can finally speak.

Step 7: learn to break the myths



« There is risk of false accusations »

2 to 10 % at the very most
vs
60 % of gender-harassment?

« We are discriminating white men »

No, we are making things fair!
(did the french revolution
« discriminate » the nobles?)

« We can't say anything anymore »

It is the converse.
Most of « we » were never
able to say anything,
and can finally speak.

« Quotas are unfair and break merit based hiring »

There is not merit currently...
(yeah, I won a race against people
carrying a 50kg backpack...)

Step 7: break the myths



« It is just a joke »



Step 7: break the myths

Act



h?



Step 7: break the myths

It is NEVER just a joke :

« sexual coercion never took place without unwanted sexual attention and gender harassment. »

(Benya, Widnall, and Johnson 2018)

Act





Step 7: break the myths

It is NEVER just a joke :

« sexual coercion never took place without unwanted sexual attention and gender harassment. »

(Benya, Widnall, and Johnson 2018)

Each layer enables the people doing worse at the next level.

Act





Step 7: break the myths

It is NEVER just a joke :

« sexual coercion never took place without unwanted sexual attention and gender harassment. »

(Benya, Widnall, and Johnson 2018)

Each layer enables the people doing worse at the next level.

Who are you laughing with???

Act





Going further
at all levels, we can act, question, investigate, and do better



When advising

- Follow **mentoring trainings**, that do take into account power imbalance.
- Favor **co-supervision** rather than one-to-one.
- **Give resources** to your students, on possible careers, local listeners.
- Be early on feedback and **don't impose any crunch**.
- **Feel responsible** (wich you often legally are) for the wellbeing of your students.
- **Emancipate** students, **don't exploit** them.
- **Never forget the factual and huge power imbalance**

→ read the booklet from <https://not-in-our-labs.github.io/>

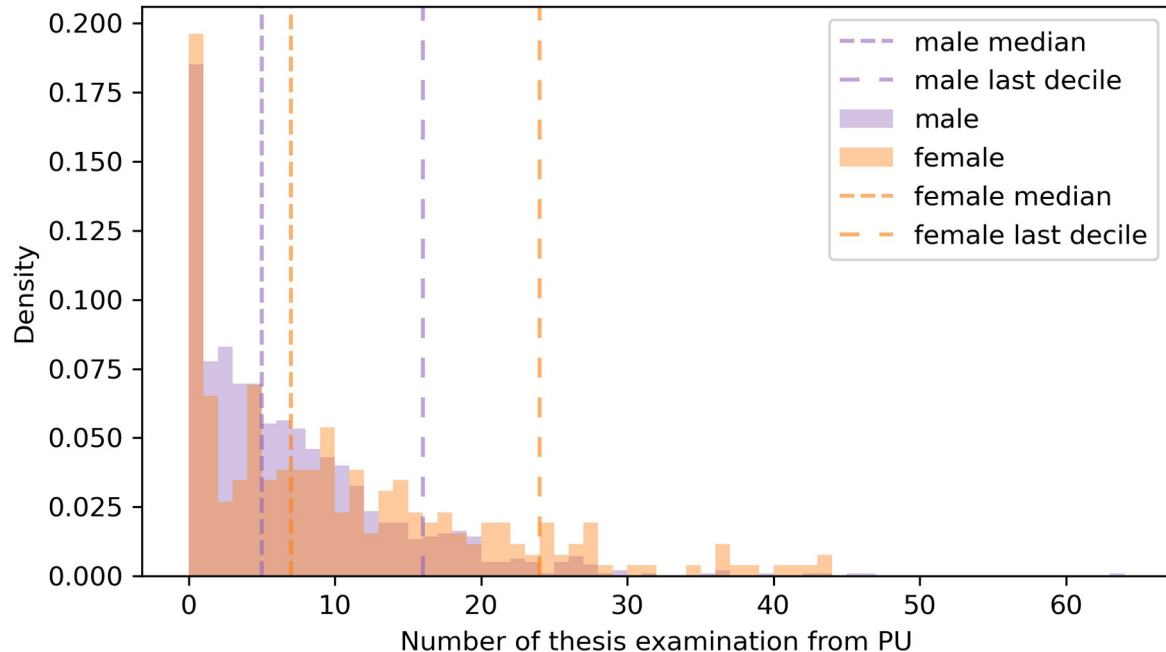
When advising



When it is time to make a
PhD thesis committe

Density function for number of thesis examination from PU per person
Section cnu (27)

Dataset of 977 male vs 260 female (21% females), France, 2015 to 2025
Total of 6569 male thesis examination from PU vs 2577 female (28% females)
Anderson-Darling test with pvalue 0.00100
Female average 9.91, male average 6.72, f-h normalized difference : 43.1%



Move things forward



When advising

**When it is time to make a
PhD thesis committee**

Don't pressure women, see mails to avoid:

<https://egalite-fh.irisa.fr/presentation-recommandations/comites-de-selection-combien-dinvitations-accepter/>

Move things forward



When advising

When it is time to make a PhD thesis committee

Don't pressure women, see mails to avoid:

<https://egalite-fh.irisa.fr/presentation-recommandations/comites-de-selection-combien-dinvitations-accepter/>

Idea? - always include in your invitation mails for women : « If you have more than 2 or 3 thesis committees already planned this year, please decline this invitation. »

Move things forward



When having responsibilities - lab heads

- Frequent **climate surveys** update
- Include **within** the usual scientific seminar talks on this
(if in need of inspiration
<https://egalite-fh.irisa.fr/animation/seminaire-fifty-fifty/>)
- Impose mentoring training, diversity workshop, bystander training
- Reduce hierarchical structures, **diffuse power**
- Encourage **dedicated discussion groups** (women-only, LGBT groups,...)



When hiring

Basic bias training with
implicit-association training
impact is not enough!

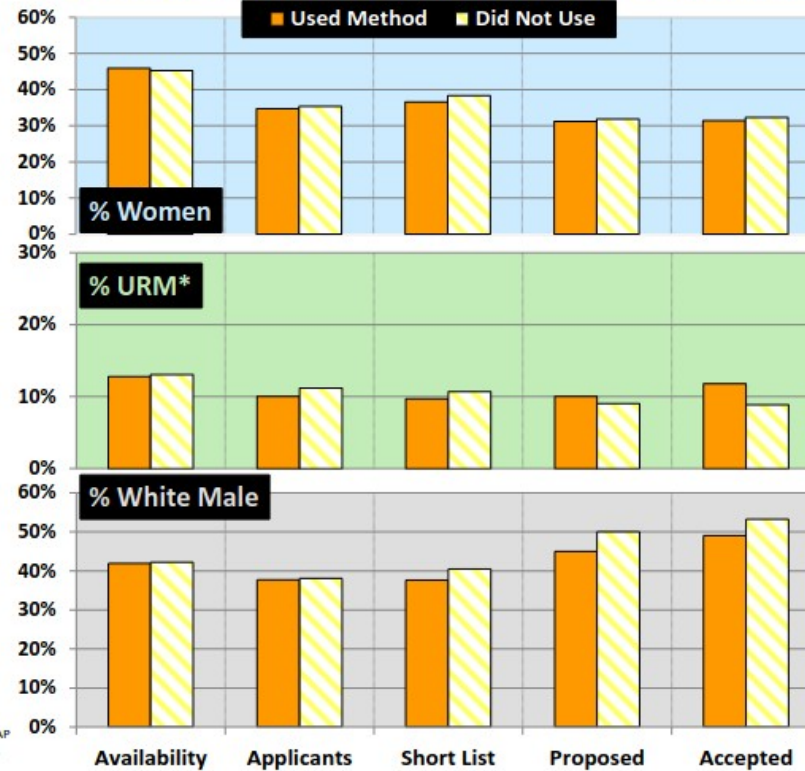
Move things forward



When hiring

Basic bias training with
implicit-association training
impact is not enough!

Figure 20a: Supported faculty peer presentations...and discussions with hiring committees regarding faculty diversity and mitigating implicit associations.



Sources: Survey of Earned Doctorates (US, PR); UCB AP Recruit 2012-13—2015-16 (as of 9/28/2016).

*includes Afric. Am.,
Hispanic, & Nat. Am.

of job searches: used
meth., N=70; partial use,
N=20; Did not use, N=130.

Move things forward



When hiring

How to do good research?



2017 → 2022
38% → 50%

Move things forward



When hiring

Recruitment measures:

- clear goals in terms of diversity
- diversification of team leaders
- 2 day training on diversity for everybody, every year.
- minimal quota of 50% of women in postdoc hiring committees
- minimal quota of 50% of women on final postdoc listings

Move things forward

How to do good research?



2017 → 2022
38% → 50%

How to do good research?



W



Recru

-
- m
lis

22
0%

Move things forward

How to do good research?



A cascading effect :

more leader women → more postdoc women → more PhD women

Recru



22
0%

Move things forward

How to do good research?



A cascading effect :

more leader women → more postdoc women → more PhD women

Recru

It is important to recruit AND keep people, and women stay more in women lead teams.

Toxic culture vs inclusive culture

22
0%

Move things forward



When hiring – so many things to try and explore!



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against **Deschamps 2024**)



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against **Deschamps 2024**)
- have clear goals and **general agreement on diversity** and discriminations. (**Goulden et al. 2019**)



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against **Deschamps 2024**)
- have clear goals and **general agreement on diversity** and discriminations. (**Goulden et al. 2019**)
- when possible, shape the job description to target diversity (**Goulden et al. 2019**)



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against [Deschamps 2024](#))
- have clear goals and **general agreement on diversity** and discriminations. ([Goulden et al. 2019](#))
- when possible, shape the job description to target diversity ([Goulden et al. 2019](#))
- ask for **CVs without dates**, only experience lengths ([Kristal et al. 2023](#)) ([Behavioural Insights Team 2021](#))



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against [Deschamps 2024](#))
- have clear goals and **general agreement on diversity** and discriminations. ([Goulden et al. 2019](#))
- when possible, shape the job description to target diversity ([Goulden et al. 2019](#))
- ask for **CVs without dates**, only experience lengths ([Kristal et al. 2023](#)) ([Behavioural Insights Team 2021](#))
- agree before hands on criterion and use **rubrics** (but be careful, a rubric can in itself discriminate, if based on something unequally easy to do) ([Culpepper et al. 2023](#)) ([Behavioural Insights Team 2021](#))



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against [Deschamps 2024](#))
- have clear goals and **general agreement on diversity** and discriminations. ([Goulden et al. 2019](#))
- when possible, shape the job description to target diversity ([Goulden et al. 2019](#))
- ask for **CVs without dates**, only experience lengths ([Kristal et al. 2023](#)) ([Behavioural Insights Team 2021](#))
- agree before hands on criterion and use **rubrics** (but be careful, a rubric can in itself discriminate, if based on something unequally easy to do) ([Culpepper et al. 2023](#)) ([Behavioural Insights Team 2021](#))
- have a jury member be responsible for highlightings whenever a new criterion appears or disappear.



When hiring – so many things to try and explore!

- have a **women for president**, but a men for the administrative part (to fight against Deschamps 2024)
- have clear goals and **general agreement on diversity** and discriminations. (Goulden et al. 2019)
- when possible, shape the job description to target diversity (Goulden et al. 2019)
- ask for **CVs without dates**, only experience lengths (Kristal et al. 2023) (Behavioural Insights Team 2021)
- agree before hands on criterion and use **rubrics** (but be careful, a rubric can in itself discriminate, if based on something unequally easy to do) (Culpepper et al. 2023) (Behavioural Insights Team 2021)
- have a jury member be responsible for highlightings whenever a new criterion appears or disappear.
- evaluate **speech time repartition** in jurys? (<http://arementalkingtoomuch.com/>) It is an objective feedback that everybody can take home and reflect on, but also a lacking data point.



When hiring – so many things to try and explore!

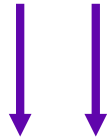
- have a **women for president**, but a men for the administrative part (to fight against Deschamps 2024)
- have clear goals and **general agreement on diversity** and discriminations. (Goulden et al. 2019)
- when possible, shape the job description to target diversity (Goulden et al. 2019)
- ask for **CVs without dates**, only experience lengths (Kristal et al. 2023) (Behavioural Insights Team 2021)
- agree before hands on criterion and use **rubrics** (but be careful, a rubric can in itself discriminate, if based on something unequally easy to do) (Culpepper et al. 2023) (Behavioural Insights Team 2021)
- have a jury member be responsible for highlightings whenever a new criterion appears or disappear.
- evaluate **speech time repartition** in jurys? (<http://arementalkingtoomuch.com/>) It is an objective feedback that everybody can take home and reflect on, but also a lacking data point.
- for people who will be teaching, include a question on what they would do in front of a discrimination: 'A bachelor male student does not want to do a group project with a female, stating that women are bad at computer science, how do you answer them?' (aiming for skill-based evaluation and awareness of those issue (Behavioural Insights Team 2021))



Time to conclude

Terrible things are here, all around us.

Competition, Star Culture, Power imbalance
Masculinist culture



Harassment, Unwell-being, Discriminations

Move things forward



Time to conclude

Terrible things are here, all around us.

Competition, Star Culture, Power imbalance
Masculinist culture

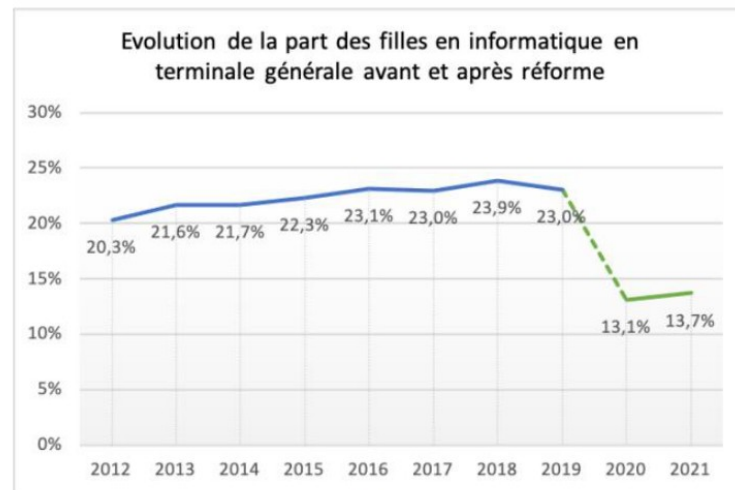


Harassment, Unwell-being, Discriminations

Move things forward

How to do good research?

It will probably worsen if we don't act



4. Part des filles en terminale



Time to conclude

Terrible things are here, all around us.

Competition, Star Culture, Power imbalance
Masculinist culture



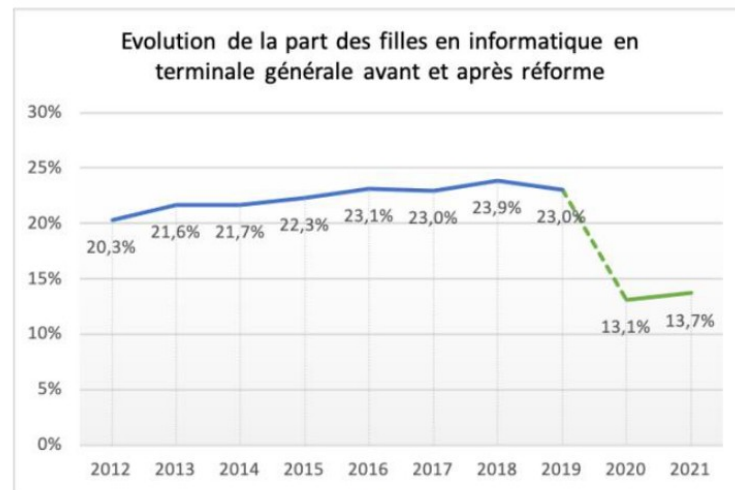
Harassment, Unwell-being, Discriminations

But, we are researchers. So, let's be researchers. Let's solve this problem!

Move things forward

How to do good research?

It will probably worsen if we don't act



4. Part des filles en terminale